

Information Retrieval

1. Introduction

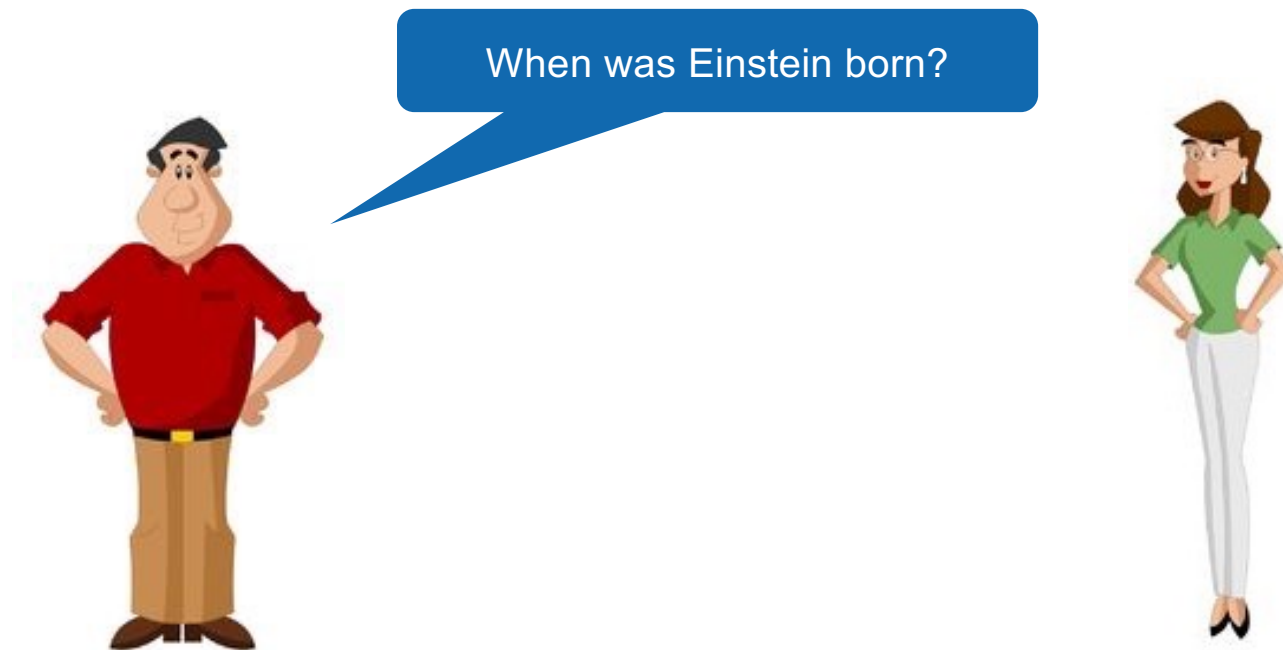
Ghislain Fourny
Spring 2022



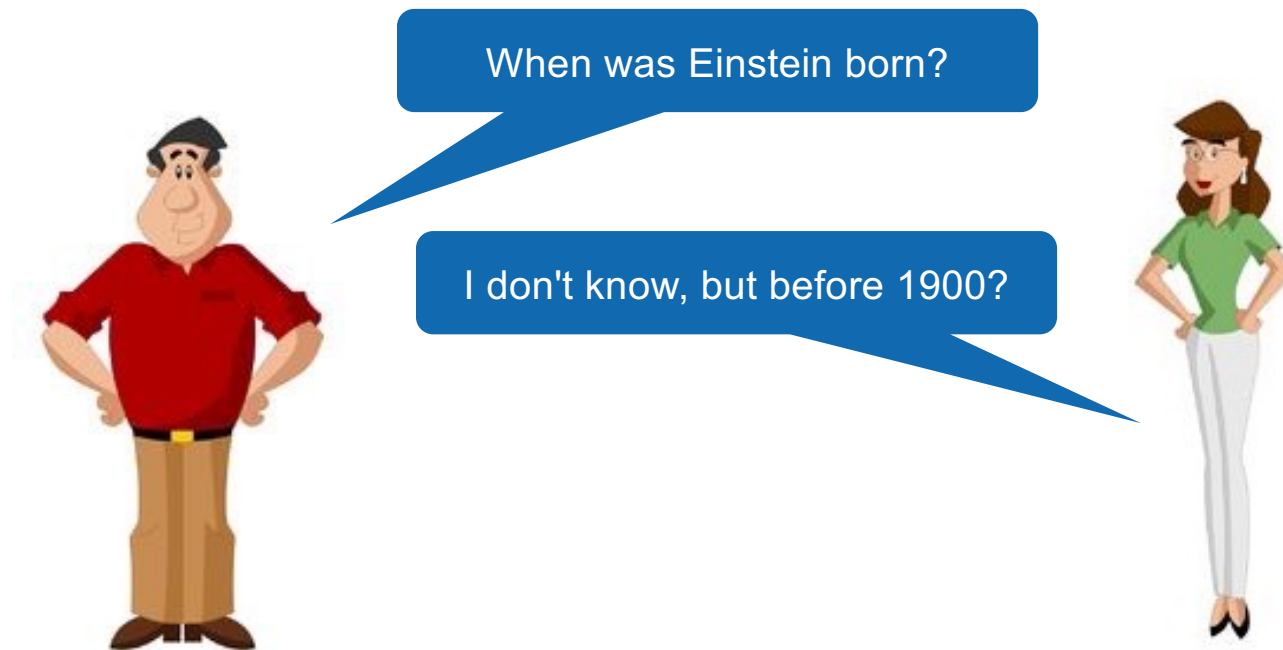
Searching for information



Searching for information



Searching for information



Poll

Go *now* to:

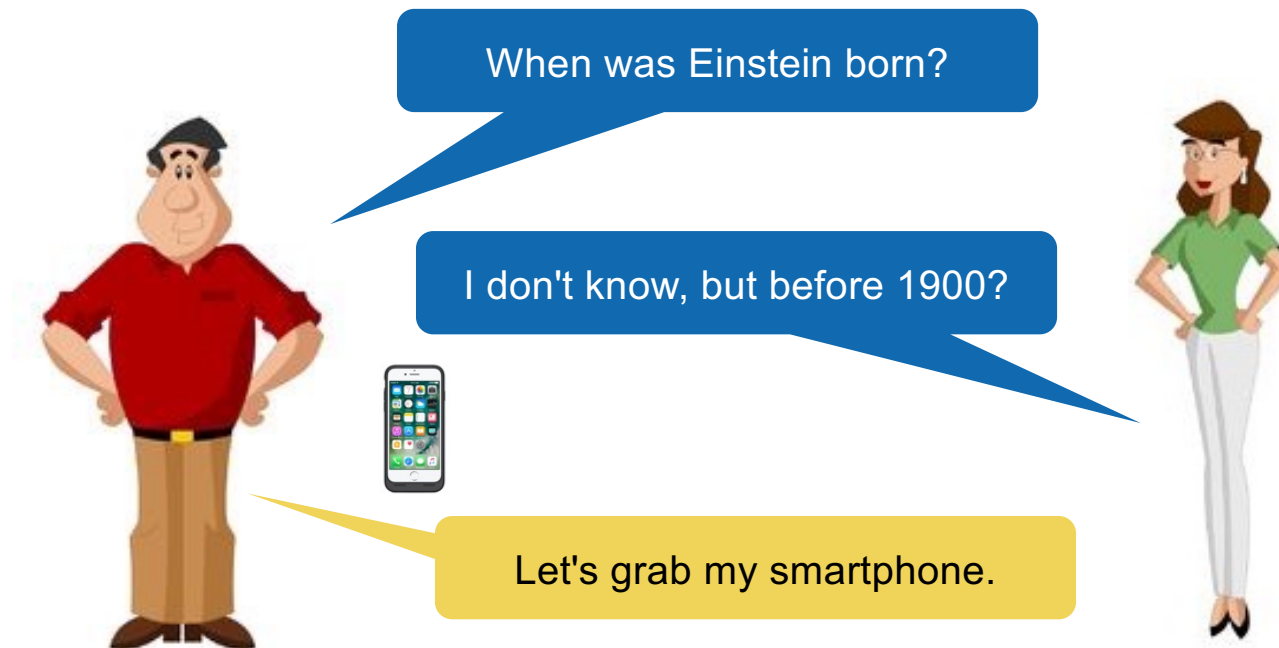
<https://eduapp-app1.ethz.ch/>



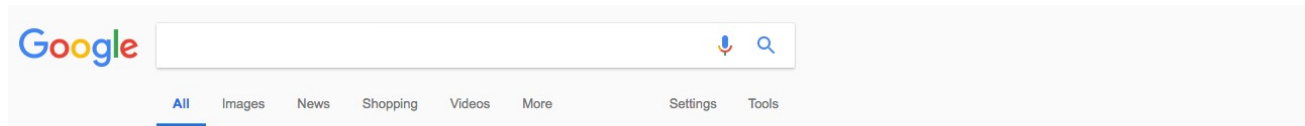
or install EduApp 3.x



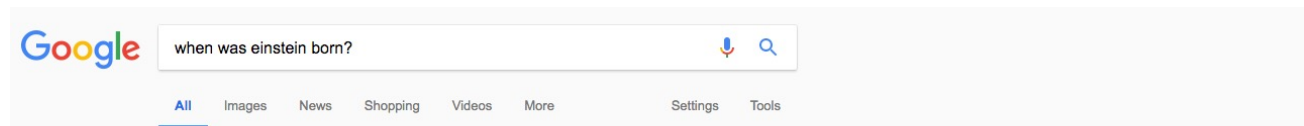
Searching for information



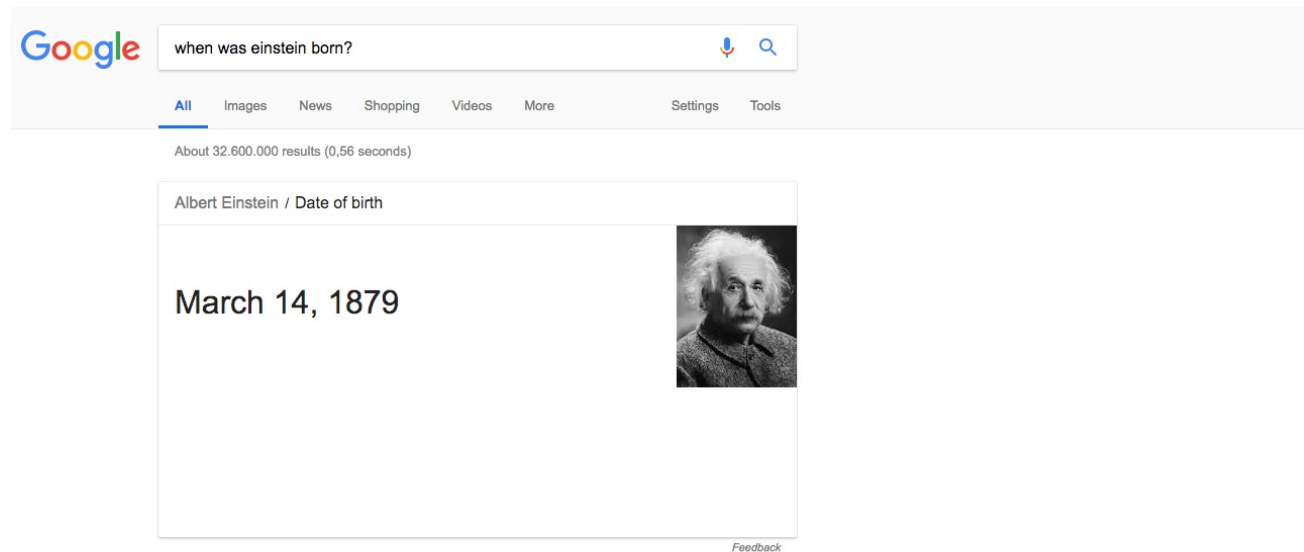
Searching for information



Searching for information



Searching for information



Searching for information

Google

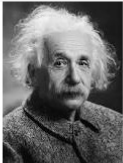
when was einstein born?

All Images News Shopping Videos More Settings Tools

About 32.600.000 results (0,56 seconds)

Albert Einstein / Date of birth

March 14, 1879



Feedback

Albert Einstein

Theoretical physicist

Albert Einstein was a German-born theoretical physicist who developed the theory of relativity, one of the two pillars of modern physics. His work is also known for its influence on the philosophy of science. [Wikipedia](#)

Born: March 14, 1879, Ulm, Germany
Died: April 18, 1955, Princeton, New Jersey, United States
Education: [University of Zurich](#) (1905), [ETH Zurich](#) (1896–1900), [MORE](#)
Spouse: [Elsa Einstein](#) (m. 1919–1936), [Mileva Marić](#) (m. 1903–1919)
Siblings: [Maja Einstein](#)

Quotes View 7+ more

Imagination is more important than knowledge.

No problem can be solved from the same level of consciousness that created it.

Life is like riding a bicycle. To keep your balance you must keep moving.

Feedback

Searching for information

Google

when was einstein born?

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About 32.600.000 results (0,56 seconds)

Albert Einstein / Date of birth

March 14, 1879

People also search for

Isaac Newton January 4, 1643

Stephen Hawking January 8, 1942

Feedback

People also ask

What did Albert Einstein invent and discover? ▾

How did Albert Einstein get famous? ▾

Where did Albert Einstein create? ▾

How old is Albert Einstein right now? ▾

Feedback

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Siblings: Maja Einstein

View 7+ more

Quotes

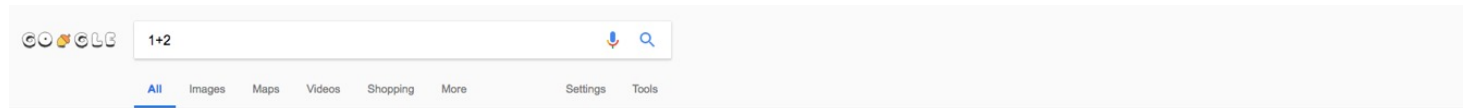
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No problem can be solved from the same level of consciousness that created it.

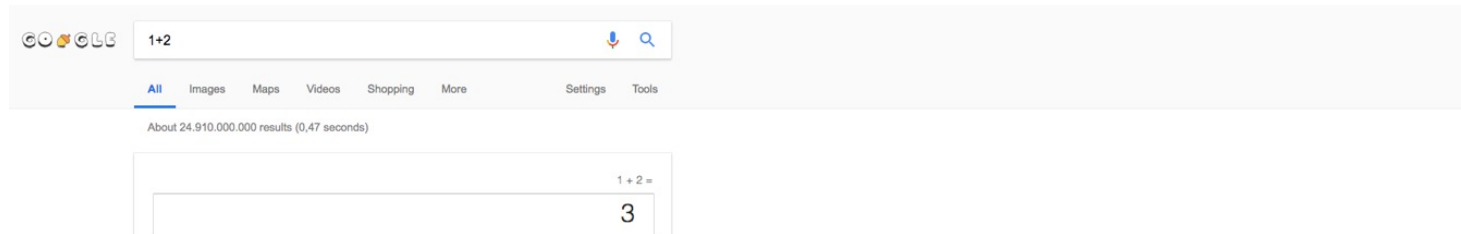
Life is like riding a bicycle. To keep your balance you must keep moving.

Feedback

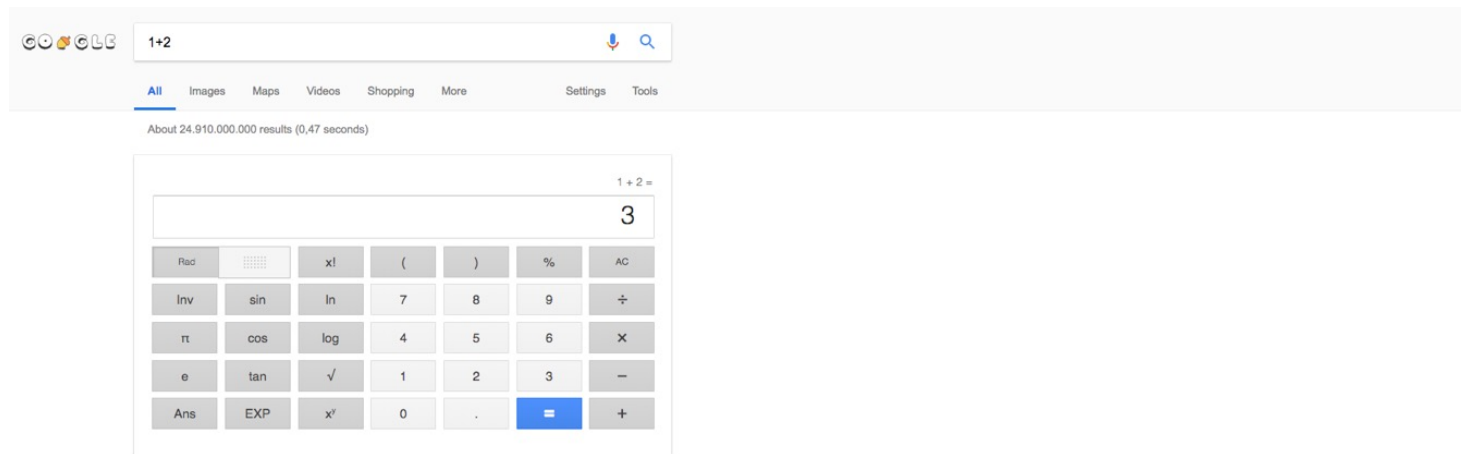
Searching for information



Searching for information



Searching for information



Searching for information

The screenshot shows a Google search interface. The search bar contains '1+2'. Below the search bar, there are tabs for 'All', 'Images', 'Maps', 'Videos', 'Shopping', 'More', 'Settings', and 'Tools'. The search results show 'About 24,910,000,000 results (0,47 seconds)'. On the left, there is a calculator widget showing '1 + 2 =' and the result '3'. On the right, there is a search result for 'Ranma 1/2' (Manga series). The result includes a thumbnail image of the main characters, a brief description, and metadata such as author, creator, books, genres, studios, and publishers. Below the main result, there is a section 'People also search for' with thumbnails for 'Inuyasha', 'Maison Ikkoku', 'Rin-ne', 'Sailor Moon', and 'Cardcaptor Sakura'.

1+2

All Images Maps Videos Shopping More Settings Tools

About 24,910,000,000 results (0,47 seconds)

1 + 2 = 3

Ranma 1/2
Manga series

Ranma 1/2 is a Japanese manga series written and illustrated by Rumiko Takahashi. It was serialized in Weekly Shōnen Sunday from September 1987 to March 1996, with the chapters collected into 38 tankōbon volumes by Shogakukan. [Wikipedia](#)

Author: Rumiko Takahashi
Creator: Rumiko Takahashi
Books: [Ranma 1/2](#), [Ranma 1/2, Vol. 26](#), [Ranma 1/2, Vol. 23](#), [MORE](#)
Genres: Comedy, Martial Arts Film
Studios: Studio Deen, Nikkatsu
Publishers: Madman Entertainment (AUS), Viz Media (NA), Shogakukan

People also search for [View 15+ more](#)

[Inuyasha](#) [Maison Ikkoku](#) [Rin-ne](#) [Sailor Moon](#) [Cardcaptor Sakura](#)

[Feedback](#)

Searching for information

1+2

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Settings Tools

About 24,910,000,000 results (0,47 seconds)

1 + 2 =

3

Rad		x!	(	)	%	AC
Inv	sin	ln	7	8	9	÷
π	cos	log	4	5	6	×
e	tan	√	1	2	3	-
Ans	EXP	x ^y	0	.	=	+

More info

Ranma ½ - Wikipedia

https://en.wikipedia.org/wiki/Ranma_½

Ranma ½ (Japanese: らんま 1/2, Hepburn: Ranma Nibun-no-Ichi, pronounced Ranma One-Half) is a Japanese manga series written and illustrated by Rumiko Takahashi. It was serialized in Weekly Shōnen Sunday from September 1987 to March 1996, with the chapters collected into 38 tankōbon volumes by Shogakukan.

Season 2 · Season 1 · List of Ranma ½ episodes · List of Ranma ½ characters

1/2 - Wikipedia

<https://en.wikipedia.org/wiki/1/2>

1/2 may refer to: One half (1/2); January 2 (month-day date notation); 1 February (day-month date notation); 1st Battalion 2nd Marines; 1 shilling and 2 pence in British predecimal currency · Disambiguation icon, This disambiguation page lists articles associated with the same number. If an internal link led you here, you ...

Template:1/2 - Wikipedia

<https://en.wikipedia.org/wiki/Template:1/2>

The templates {{1/2}}, {{1/3}}, {{2/3}}, {{1/4}} and {{3/4}} use HTML markup to create large, readable, common fractions "1/2", "1/3", "2/3", "1/4", "3/4", as opposed to the often illegible "¼", "⅓", "½", "¾" or horizontal "1⁄2", "1⁄3", "2⁄3", "1⁄4", "3⁄4". They are intended as concise, easily interpreted alternatives to {{frac|1/2}}, {{frac|1/3}}, ...

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Inuyasha


Maison Ikkoku


Rin-ne


Sailor Moon

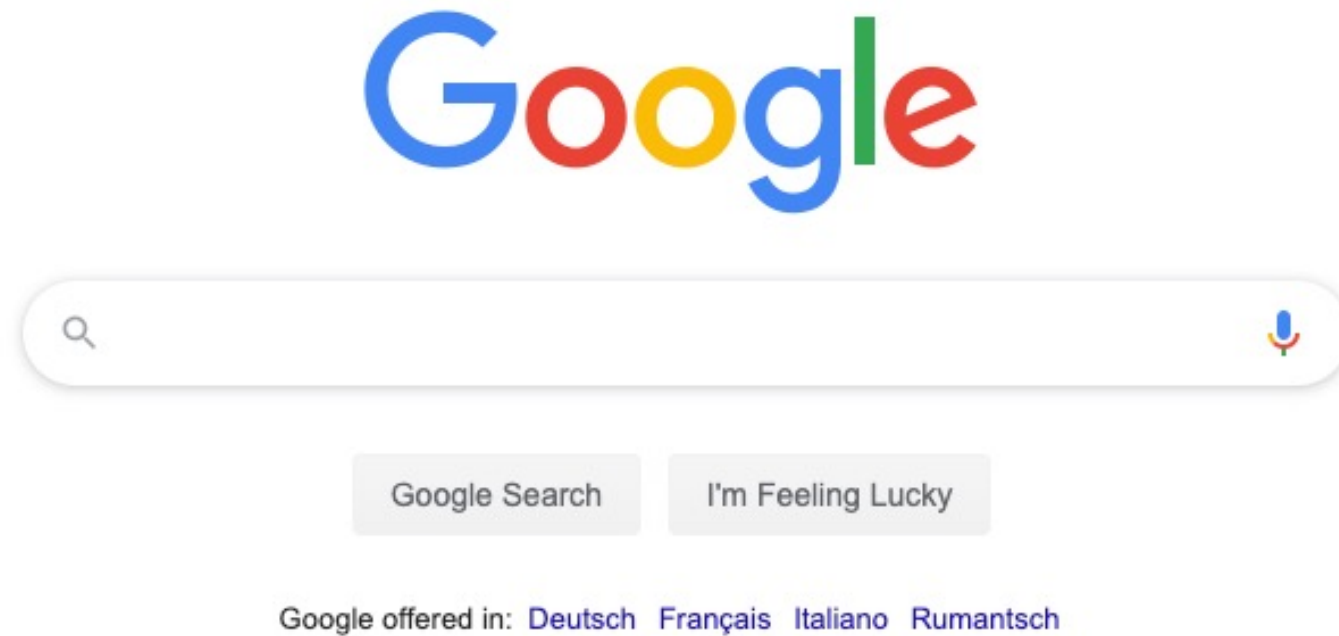

Cardcaptor Sakura

Feedback

In 1998



2019



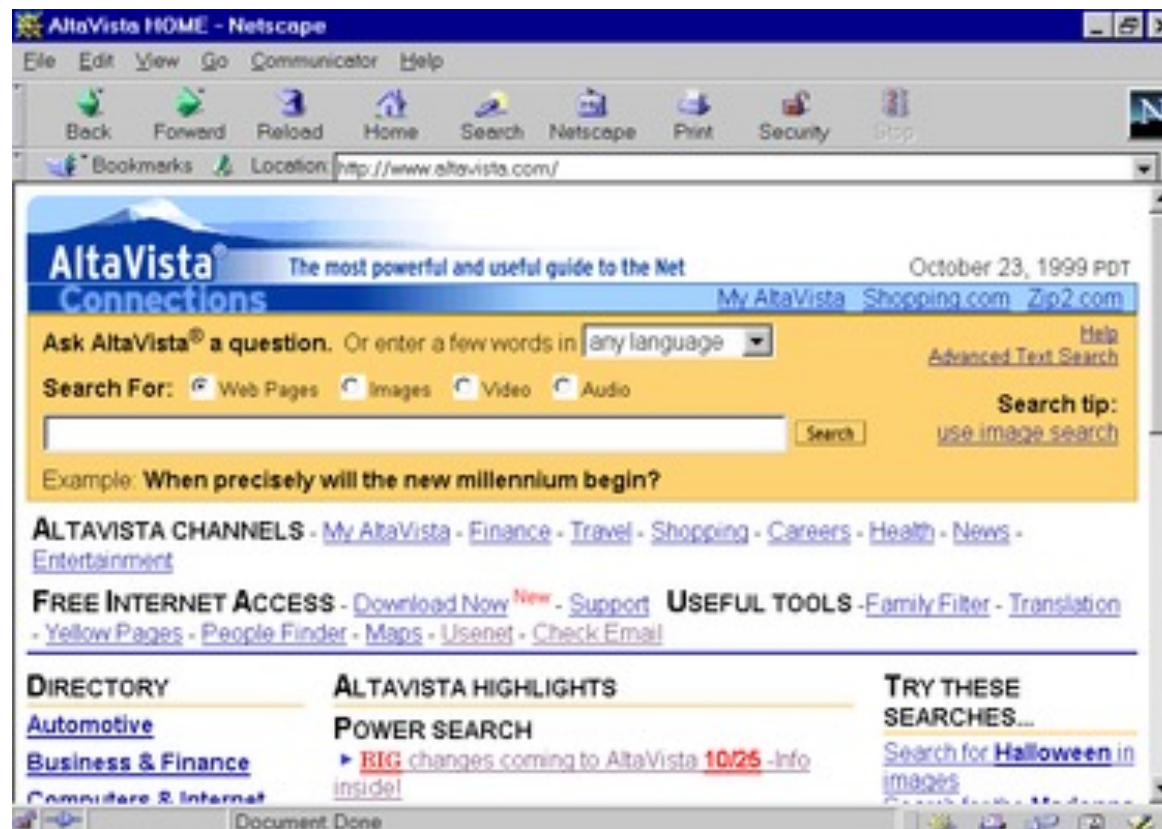
1998



1996



1996



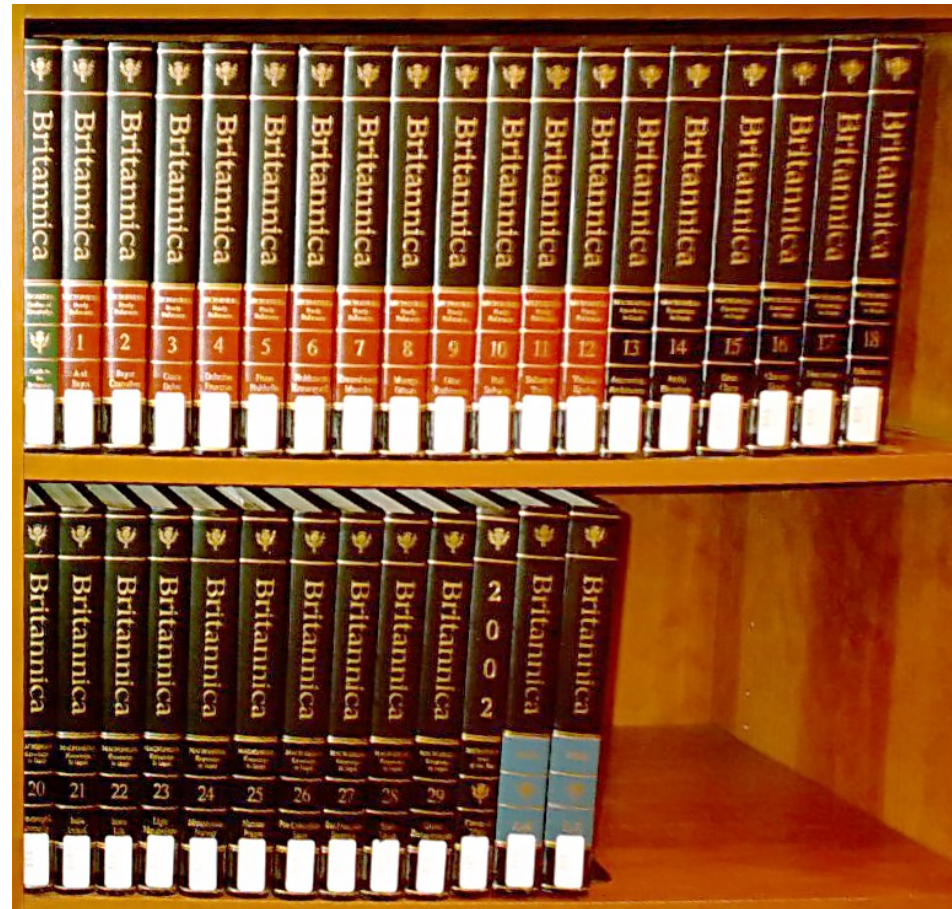
1996



1993-2009



1768-2010



Knowledge repositories

Library of Alexandria (280 BC): 40,000 – 400,000 scrolls



O. Von Corven - Tolzmann, Don Heinrich, Alfred Hessel and Reuben Peiss. *The Memory of Mankind*. New Castle, DE: Oak Knoll Press, 2001

Knowledge repositories

Vatican Library (1475): 3,600 codices

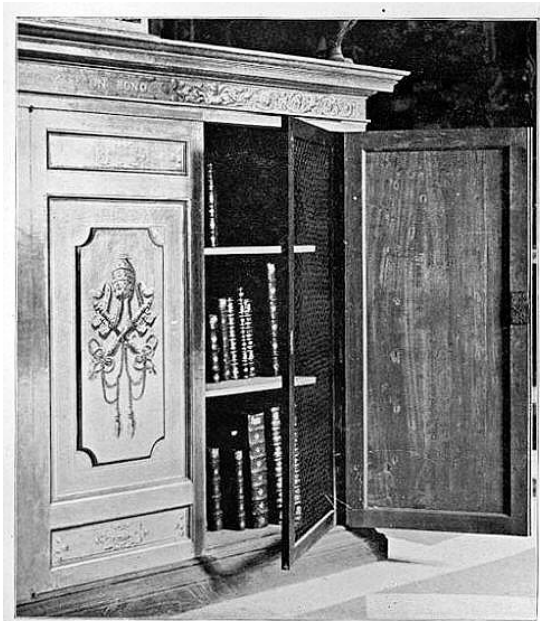


Fig. 17. A single press in the Vatican Library, open.
From a photograph.

John Willis Clark - Fritz-Milkau-Dia-Sammlung, erstellt in der Photographischen Werkstatt der Preußischen Staatsbibliothek von 1926-1933 originally from *The Care of Books* by John Willis Clark (Fig 17, p48)

Knowledge repositories

Herzog-August-Bibliothek (1661): 116,000 books



By User:Bruswyk - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=559061>

Knowledge repositories

British Museum (1845): 240,000 books



By Ham - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=1553456>

Knowledge repositories

Library of Congress (1990): 100,000,000 documents



By Carol M. Highsmith - This image is available from the United States Library of Congress's Prints and Photographs division under the digital ID highsm.11604. This tag does not indicate the copyright status of the attached work. A normal copyright tag is still required. See Commons:Licensing for more information. العربية | čeština | Deutsch | English | español | فارسی | suomi | français | עברית | magyar | italiano | македонски | עברית | Nederlands | polski | português | русский | slovenčina | slovenščina | Türkçe | українська | 中文 | 中文（简体） | 中文（繁體） | +/-, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=15709590>

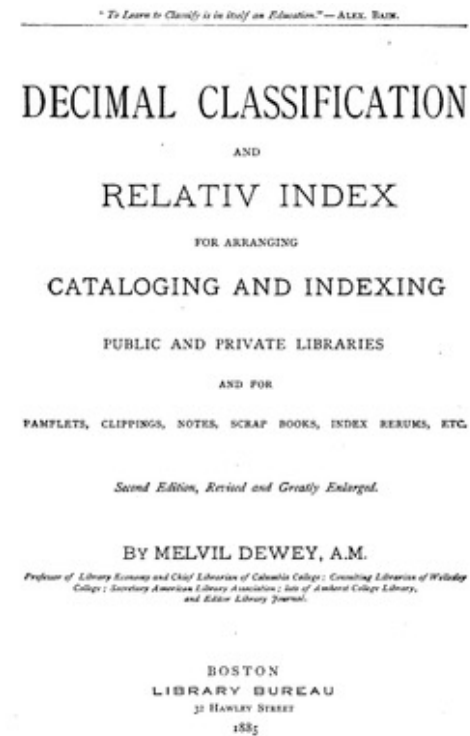
Shanghai library: today's second largest



British library: today's largest



Dewey Decimal System



Dewey Decimal System



By Skwanem - Own work, CC BY-SA 3.0,
<https://commons.wikimedia.org/w/index.php?curid=6086987>

Dewey Decimal System

000 Generalities - Data processing - Computer science - Library information and sciences - General encyclopedias - News media - Journalism	500 Natural Sciences & Math - Math - Astronomy - Physics - Chemistry - Earth sciences - Life sciences, Biology - Plants - Wild animals
100 Philosophy & Psychology - Philosophy - Paranormal phenomena (occult, witchcraft, astrology, ESP, ghosts, dreams, etc.) - Psychology - Logic - Ethics	600 Technology (Applied Sciences) - Medical sciences - Engineering - Agriculture, Farm animals & pets - Home economics & family living - Business management - Manufacturing - Buildings, Construction
200 Religion - Theory of Religion - The Bible - Christianity - Practice & Observance - Religious Orders - History & Denominations - World Religions	700 The Arts - Civic and landscape art - Architecture - Sculpture - Drawing & decorative arts - Painting - Graphic arts - Photography - Music - Performing arts - Sports, Games, Outdoor activities
300 Social Sciences - Sociology - Anthropology - Political science - Economics - Law - Military science - Education - Commerce, Communication, Transportation - Customs, Folklore	800 Literature - Literature of all languages - Poetry - Drama - Fiction - Essays - Speeches - Letters - Humor & satire
400 Language - Linguistics - World languages - Dictionaries	900 Geography, History & Biography - Geography - Travel - Biography - Ancient world - Europe - Asia - Africa - North America - South America - Extraterrestrial worlds

Dewey Number	Cute little Cave guy!	Categories	Examples
000		<i>Basic Information</i> Generalities (Information you need to learn)	Encyclopedias Newspapers Books about libraries Internet
100		<i>Who am I?</i> Philosophy & Psychology (Man thinks about himself)	Things we don't understand: UFO's Ghosts Bugfoot Aliens from outer space
200		<i>Who made me?</i> Religion (Man thinks about God)	Any religions Catholic, Jewish, Baptist, etc. Mythology
300		<i>Who lives next door?</i> Social Sciences (Man thinks about others)	Law & Government Jobs & Money Families & holidays Folk & Fairy Tales
400		<i>How can I talk to others?</i> Languages (Philology) (Man wants to communicate)	Any language French, German, Italian Dictionaries from any language Books on sign language
500		<i>What's around me?</i> Natural Science* (Man thinks about Nature)	Anything that happens by itself Rocks, trees, weather, oceans, rivers, wild animals, dinosaurs, plants, space, moon, stars, sun, math, time, chemistry
600		<i>How can I use Nature?</i> Applied Science* (Man thinks about his world)	Cars, boats, planes, cookbooks, computers, pets, farm animals, medicine, human body
700		<i>What can I do for fun?</i> Arts & Recreation (Man thinks about leisure time)	Dance, music, theater, drawing, jokes, riddles, sports <i>(One child pointed out all their "special classes" are here)</i>
800		<i>How can I record deeds?</i> Literature (Man becomes a storyteller)	Any literature: Spanish, French, English Shakespeare Poems Plays
900		<i>How can I leave a record?</i> History, Geography & Biography (Man thinks about history)	Countries, s States Biographies historical events Wars Travel books

<http://www.ais.up.ac.za/~vet/infomania/infomania14/dewey14.htm>

Verschlagwortung



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[→ Formats and Communication Interfaces](#)

[↓ Integrated Authority File \(GND\)](#)

[→ GND Change Service in MARC 21 Format](#)
[→ GND tools](#)
[→ GND Formats and Interfaces](#)

INTEGRATED AUTHORITY FILE (GND)

The Integrated Authority File (GND) is an authority file for Persons, Corporate bodies, Conferences and Events, Geographic Information, Topics and Works. It is used above all for the cataloguing of literature by libraries, but it is also increasingly being deployed in archives, museums, projects and web applications. It is operated cooperatively by the German National Library, all German-speaking library networks and the associated libraries, the German Union Catalogue of Serials ([↗ ZDB](#)) and numerous other institutions. Contributions to the GND are made either via the networks or in direct agreement with the German National Library.

In April 2012 the GND replaced the following previously separate authority files: the German Personal Name Authority File (PND), the Corporate Body Authority File (GKD), the Subject Headings Authority File (SWD) and the Uniform Title File of the Deutsches Musikarchiv (DMA-EST file). Previous format differences were overcome through the integration of all authority records in the GND. Duplicate data records from different authority files will eventually be combined in the future.

The cataloguing codes upon which the authority records are based are the "Resource Description and Access" (RDA) for the entities which are used for descriptive and subject cataloguing such as persons, and also the "Regeln für den Schlagwortkatalog" ([↗ RSWK](#)) for entities which are used exclusively for subject cataloguing, such as topics.

Authority data simplify cataloguing while providing clear retrieval access points and the possibility to link together different information resources. The GND authority records standardise the preferred names but also provide alternative names and relations to other authority records. This has resulted in the creation of a network of related data records which is especially well suited for use in the Web, permits navigation within the authority file and thereby improves search possibilities for users. In addition, some geographic authority data contain unchanged coordinates from the [↗ GeoNames](#) database.

The GND authority data are available to all interested parties in different ways, free of charge under licence [↗ CC0 1.0](#), in the following formats: MARC 21 Authority, MARC21-xml and RDFxml.

GND authority records form part of the Virtual International Authority File ([↗ VIAF](#)); they are combined with data from other national authority files.

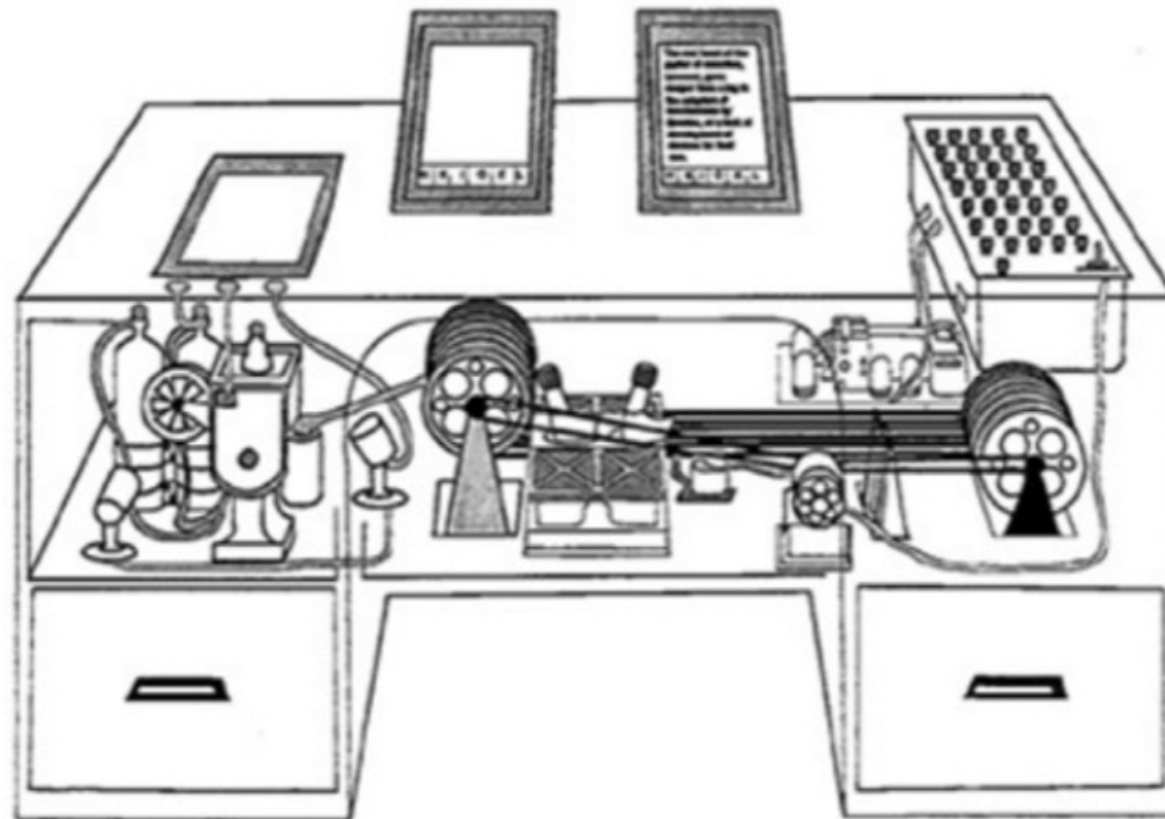
[↗ Information page on GND in the German National Library Wiki](#)

@ Normdateien

Last update: 21.10.2016
Short-URL: <http://www.dnb.de/EN/gnd>

[↑ to the top](#)

Memex



Vannevar Bush



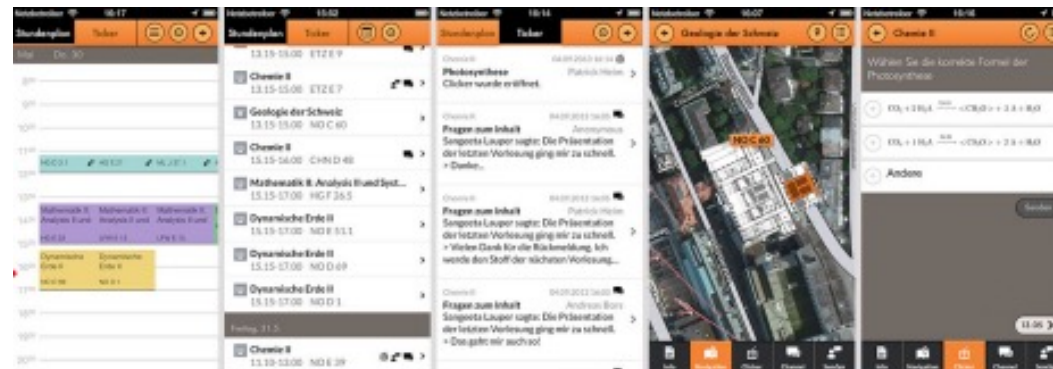
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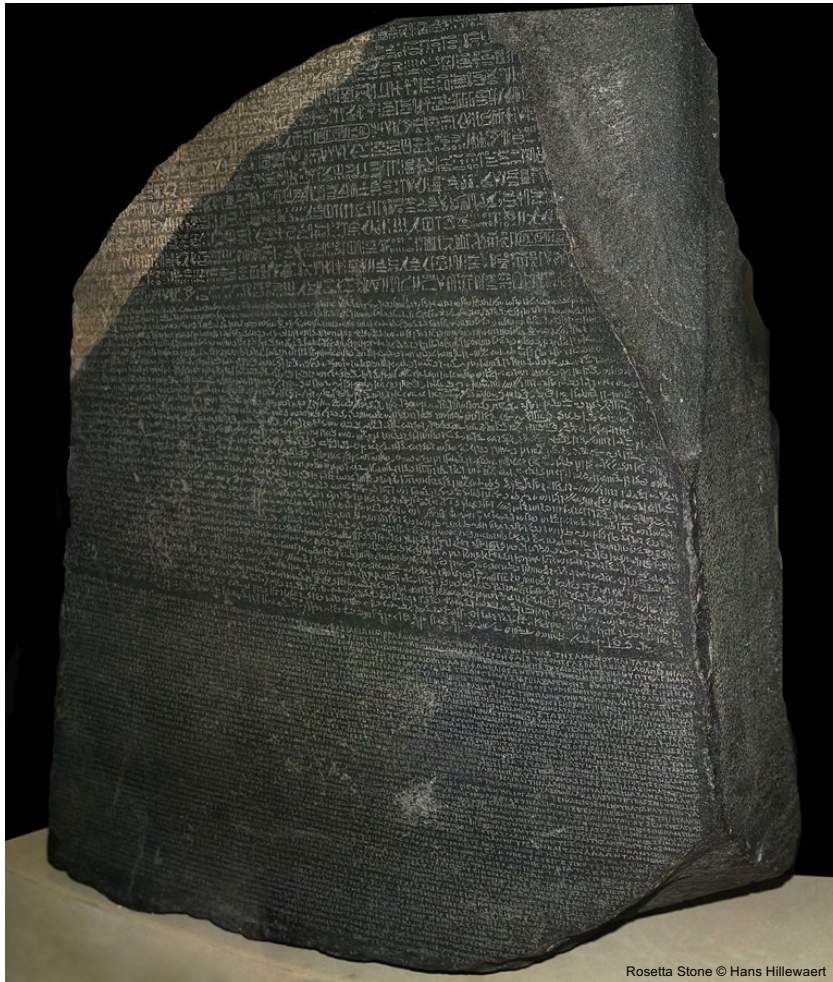


A Short History of Databases



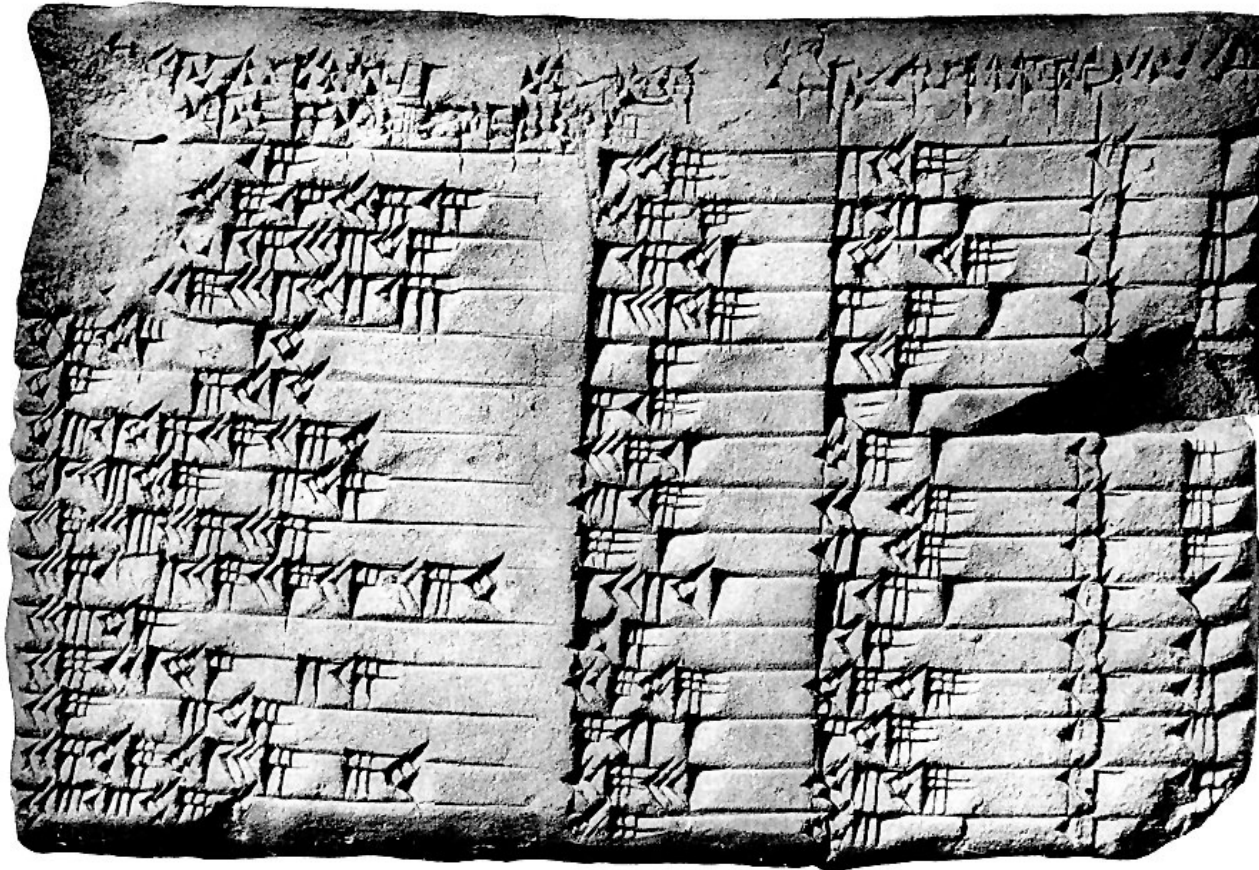


Speaking/Singing



Writing

Accounting





Willi Heidelbach

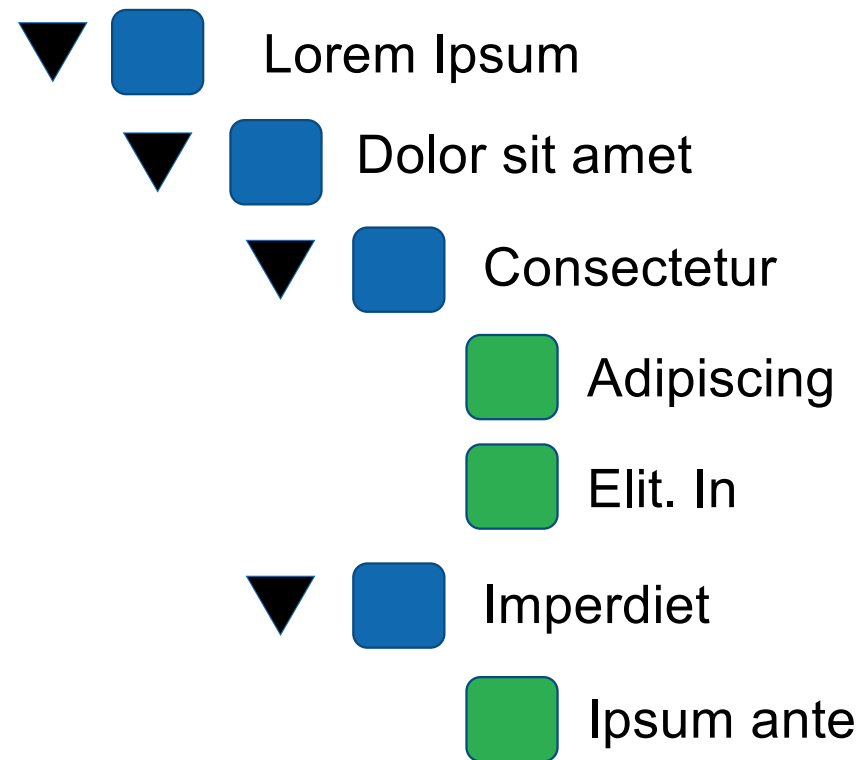
Printing



Ben Franske - DM IBM S360.jpg on en.wiki

Computers

1960s: File Systems





1970s: The Relational Era

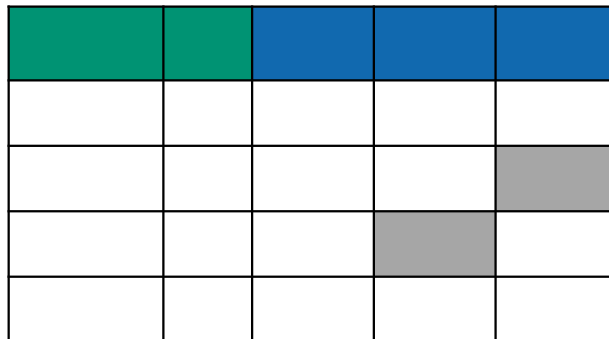
1980s: The Object Era



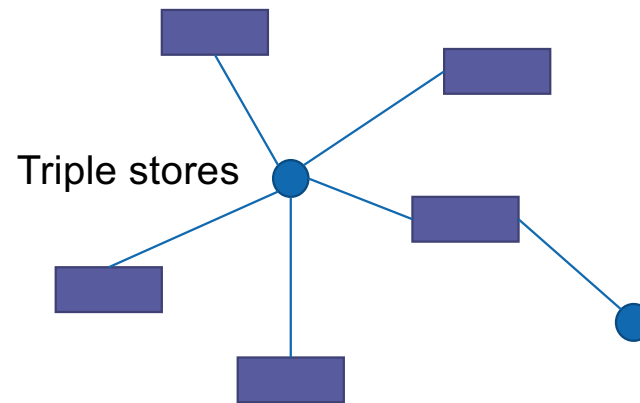
2000s: The NoSQL Era



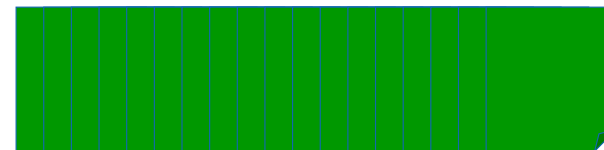
Key-value stores



Column stores

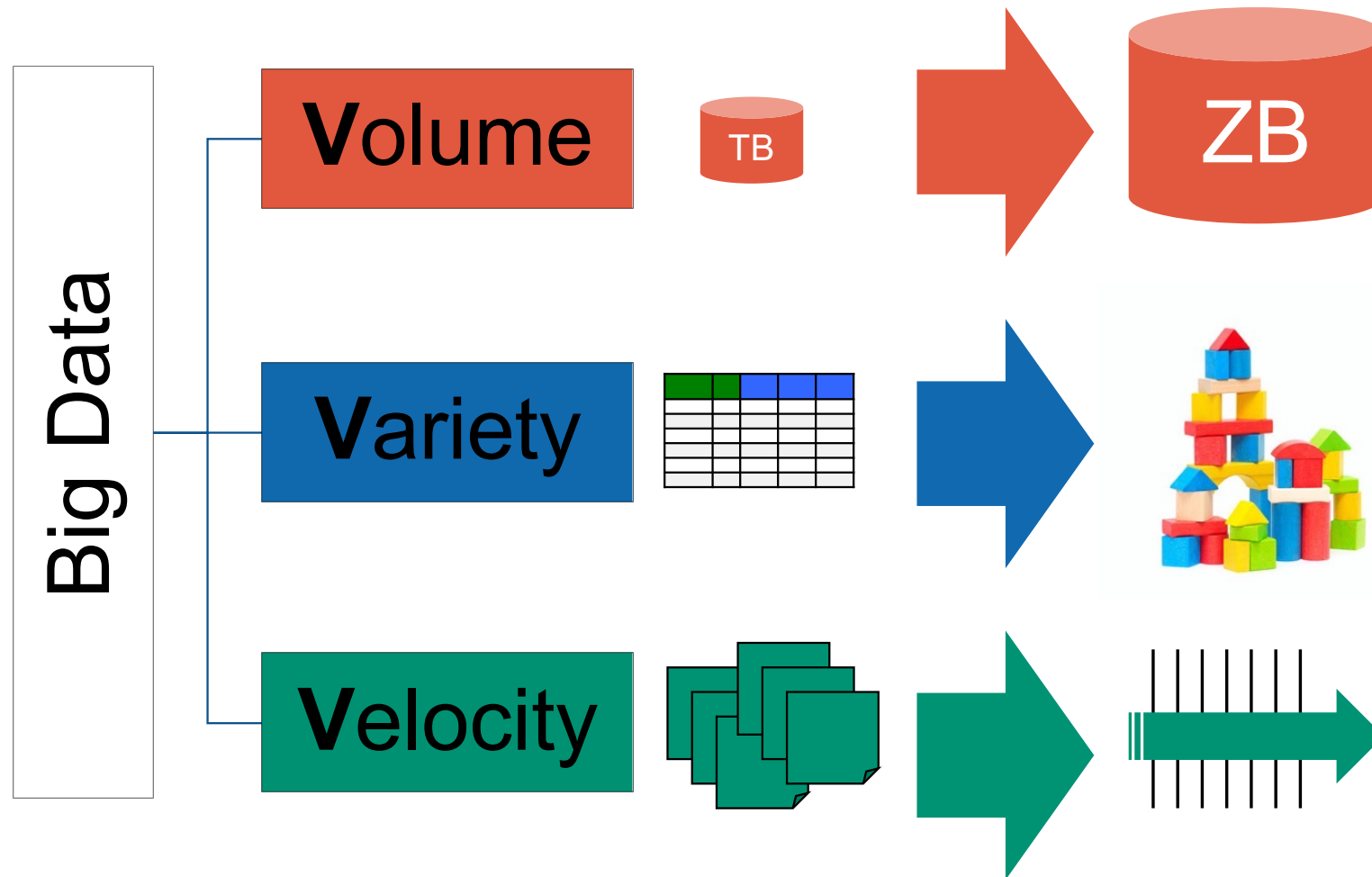


Triple stores



Document stores

The Three Vs



Prefixes (International System of Units)

kilo (k)	1,000 (3 zeros)
Mega (M)	1,000,000 (6 zeros)
Giga (G)	1,000,000,000 (9 zeros)
Tera (T)	1,000,000,000,000 (12 zeros)
Peta (P)	1,000,000,000,000,000 (15 zeros)
Exa (E)	1,000,000,000,000,000,000 (18 zeros)
Zetta (Z)	1,000,000,000,000,000,000,000 (21 zeros)
Yotta (Y)	1,000,000,000,000,000,000,000,000 (24 zeros)

You must know this by ❤️!



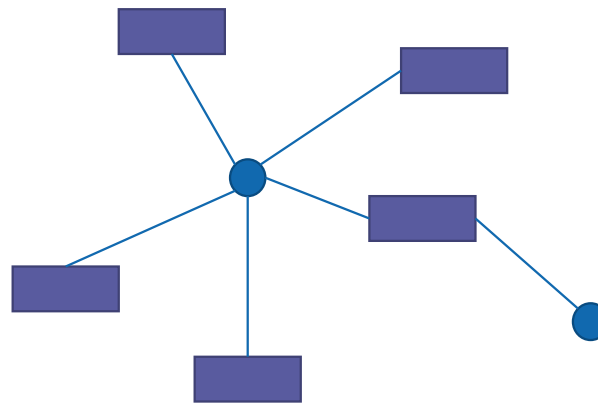
Data Variety

Data Shapes: Tables

Data Shapes: Trees



Data Shapes: Graphs



Data Shapes: Cubes

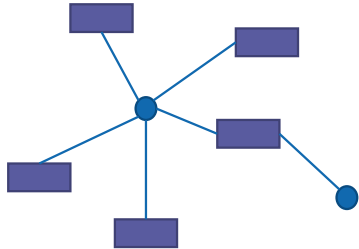


Data Shapes: Text

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Praesent nec libero metus. Praesent at turpis placerat, congue ipsum eget, scelerisque justo. Ut volutpat, massa ac lacinia cursus, nisl dui volutpat arcu, quis interdum sapien turpis in tellus. Suspendisse potenti. Vestibulum pharetra justo massa, ac venenatis mi condimentum nec. Proin viverra tortor non orci suscipit rutrum. Phasellus sit amet euismod diam. Nullam convallis nunc sit amet diam suscipit dapibus. Integer porta hendrerit nunc. Quisque pharetra congue porta. Suspendisse vestibulum sed mi in euismod. Etiam a purus suscipit, accumsan nibh vel, posuere ipsum. Nulla nec tempor nibh, id venenatis lectus. Duis lobortis id urna eget tincidunt.

Data Shapes: Text



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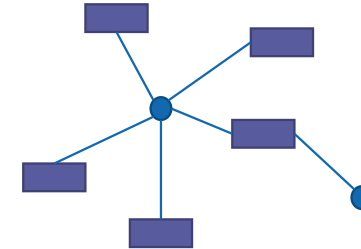
Praesent nec libero metus. Praesent at turpis placerat, congue ipsum eget, scelerisque justo. Ut volutpat, massa ac lacinia cursus, nisl dui volutpat arcu, quis interdum sapien turpis in tellus. Suspendisse potenti. Vestibulum pharetra justo massa, ac venenatis mi condimentum nec. Proin viverra tortor non orci suscipit rutrum. Phasellus sit amet euismod diam. Nullam convallis nunc sit amet diam suscipit dapibus. Integer porta hendrerit nunc. Quisque pharetra congue porta. Suspendisse vestibulum sed mi in euismod. Etiam a purus suscipit, accumsan nibh vel, posuere ipsum. Nulla nec tempor nibh, id venenatis lectus. Duis lobortis id urna eget tincidunt.



Data Shapes: Text

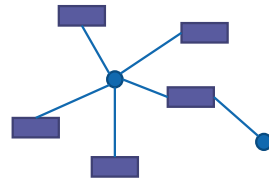
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam vel erat nec dui aliquet vulputate sed quis nulla. Donec eget ultricies magna, eu dignissim elit. Nullam sed urna nec nisl rhoncus ullamcorper placerat et enim. Integer varius ornare libero quis consequat. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean eu efficitur orci. Aenean ac posuere tellus. Ut id commodo turpis.

Praesent nec libero metus. Praesent at turpis placerat, congue ipsum eget, scelerisque justo. Ut volutpat, massa ac lacinia cursus, nisl dui volutpat arcu, quis interdum sapien turpis in tellus. Suspendisse potenti. Vestibulum pharetra justo massa, ac venenatis mi condimentum nec. Proin viverra tortor non orci suscipit rutrum. Phasellus sit amet euismod diam. Nullam convallis nunc sit amet diam suscipit dapibus. Integer porta hendrerit nunc. Quisque pharetra congue porta. Suspendisse vestibulum sed mi in euismod. Etiam a purus suscipit, accumsan nibh vel, posuere ipsum. Nulla nec tempor nibh, id venenatis lectus. Duis lobortis id urna eget tincidunt.



Data Structure

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam vel erat nec dui aliquet vulputate sed quis nulla. Donec eget ultricies magna, eu dignissim elit. Nullam sed urna nec nisl rhoncus ullamcorper placerat et enim. Integer varius ornare libero quis consequat. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aenean eu efficitur orci. Aenean ac posuere tellus. Ut id commodo turpis.



Unstructured
data

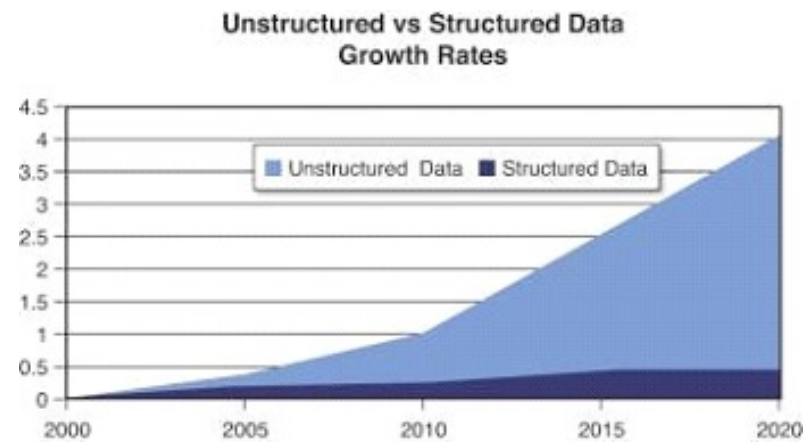
Semi-structured
data

Structured
data

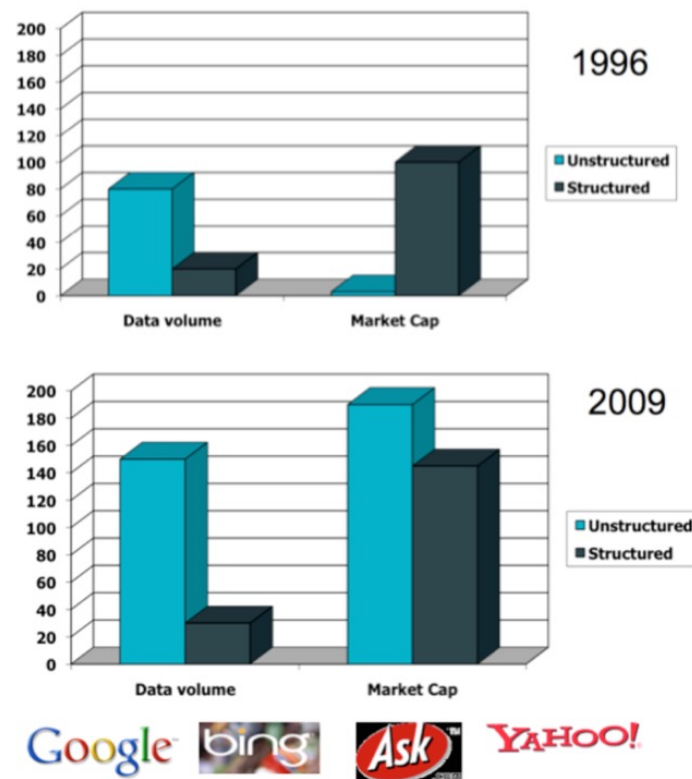


Unstructured data

The importance of unstructured data



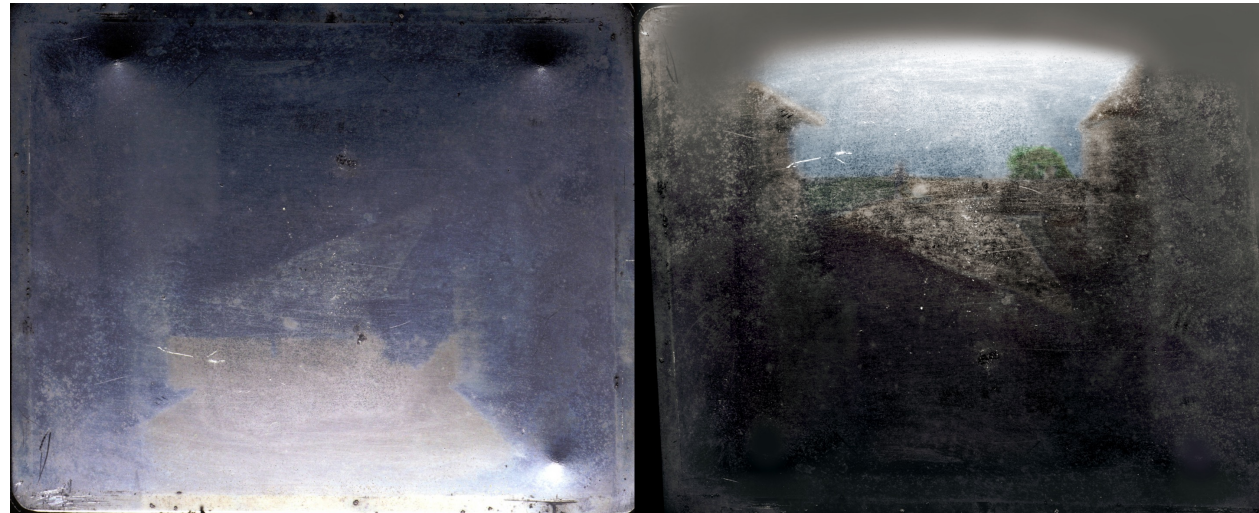
The importance of unstructured data



Unstructured data: Text

Of all the problems which have been submitted to my friend, Mr. Sherlock Holmes, for solution during the years of our intimacy, there were only two which I was the means of introducing to his notice--that of Mr. Hatherley's thumb, and that of Colonel Warburton's madness. Of these the latter may have afforded a finer field for an acute and original observer, but the other was so strange in its inception and so dramatic in its details that it may be the more worthy of being placed upon record, even if it gave my friend fewer openings for those deductive methods of reasoning by which he achieved such remarkable results. The story has, I believe, been told more than once in the newspapers, but, like all such narratives, its effect is much less striking when set forth en bloc in a single half-column of print than when the facts slowly evolve before your own eyes, and the mystery clears gradually away as each new discovery furnishes a step which leads on to the complete truth. At the time the circumstances made a deep impression upon me, and the lapse of two years has hardly served to weaken the effect.

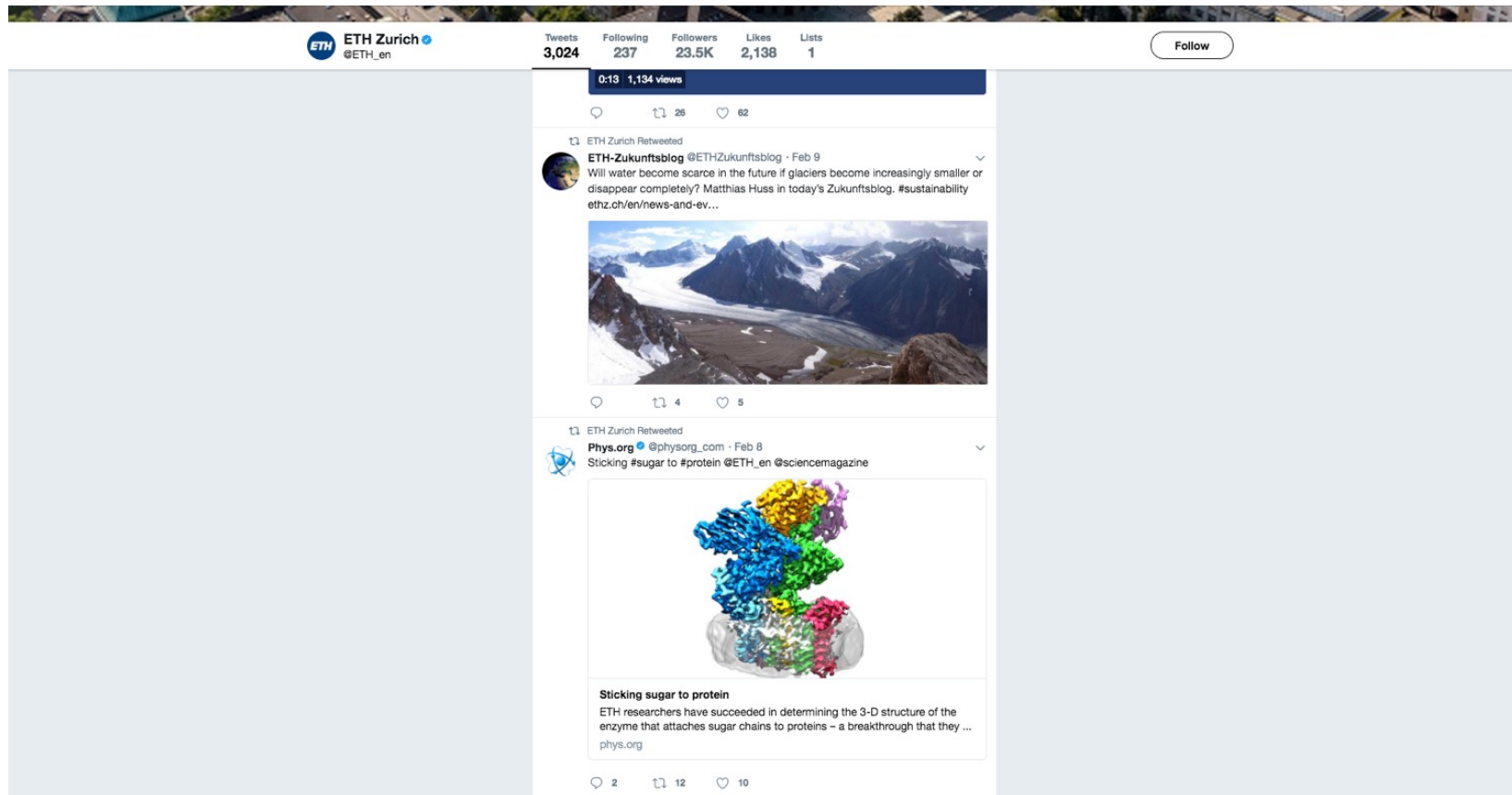
Unstructured data: Picture (bitmap)



Unstructured data: Audio



Unstructured data: Tweets



Unstructured data: Video (sequence of bitmaps)



Unstructured data

... is in a very **raw** form

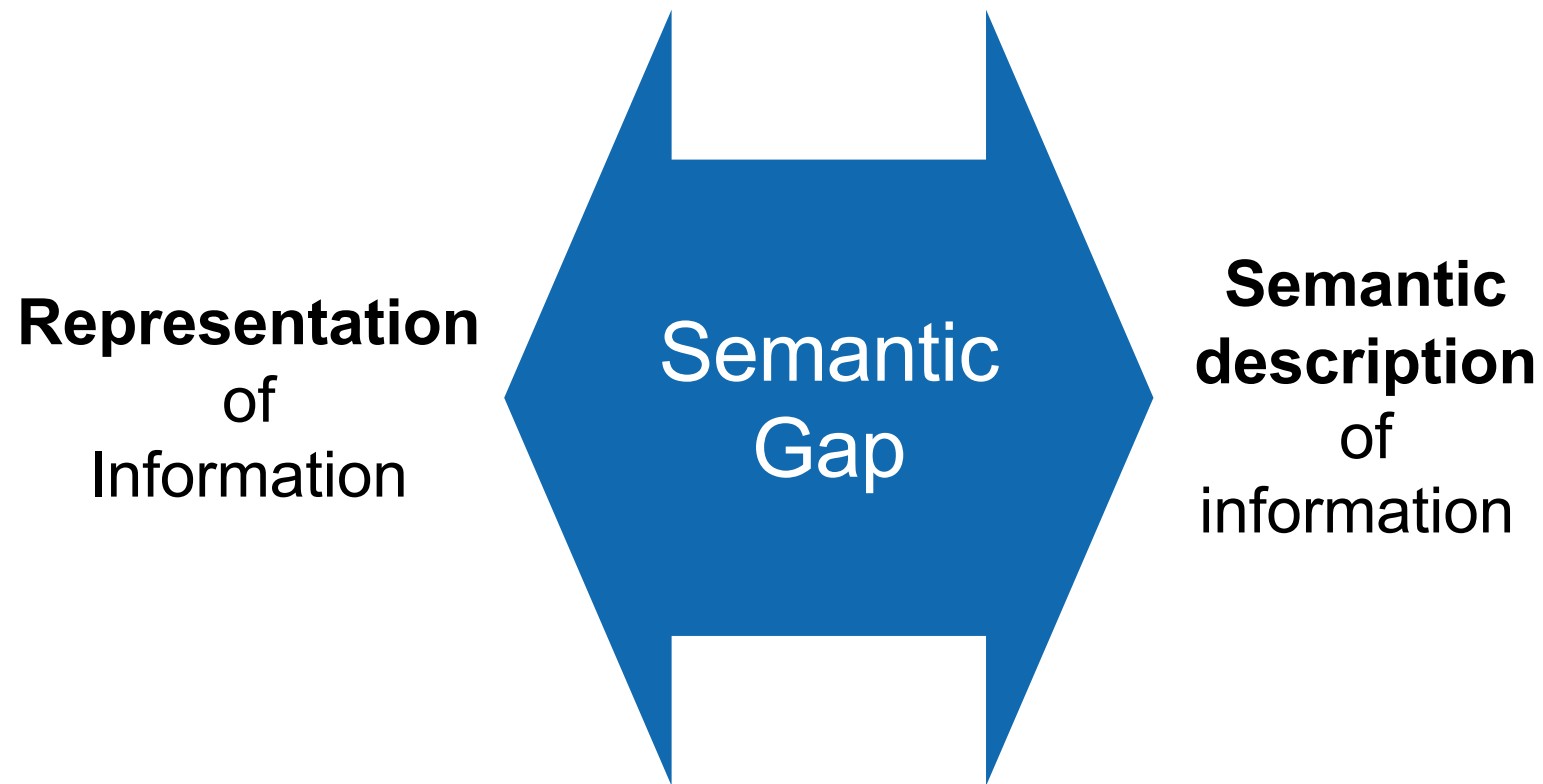


Unstructured data

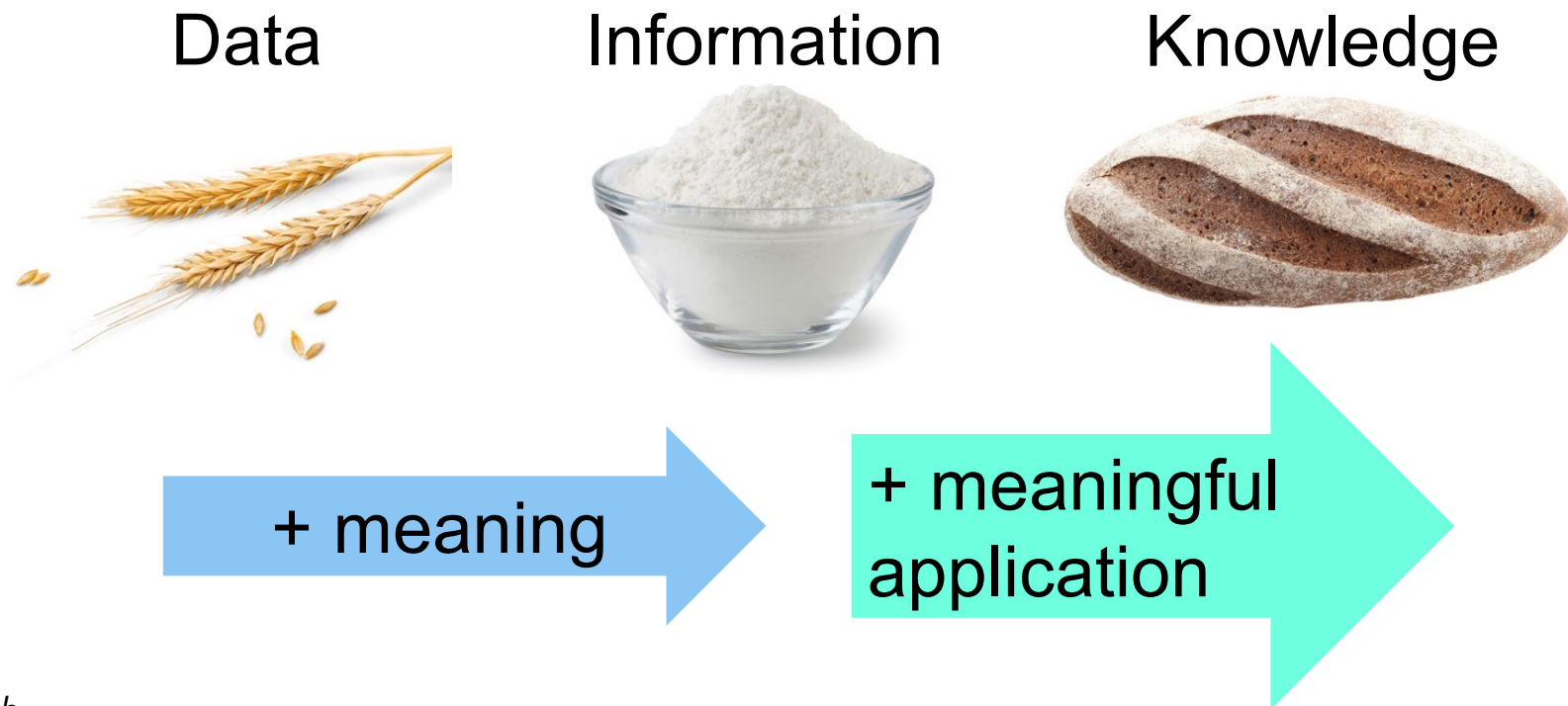
No **explicit** structure
but implicit features

(characters, grammar, pixels, waves, etc)

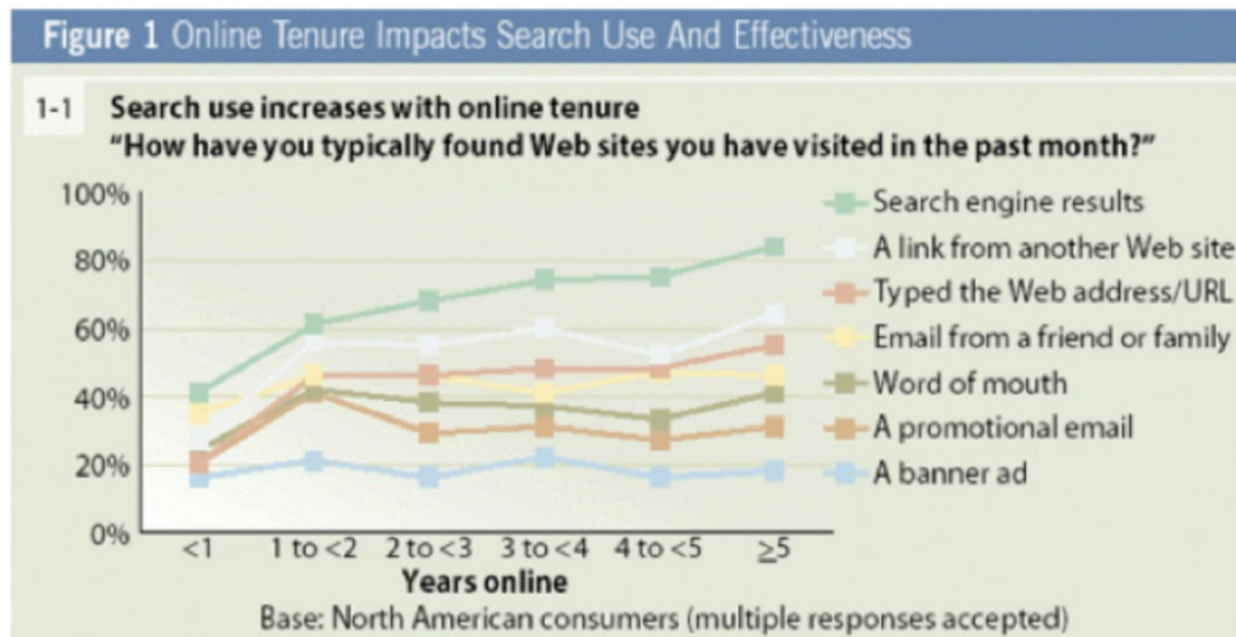
Semantic gap



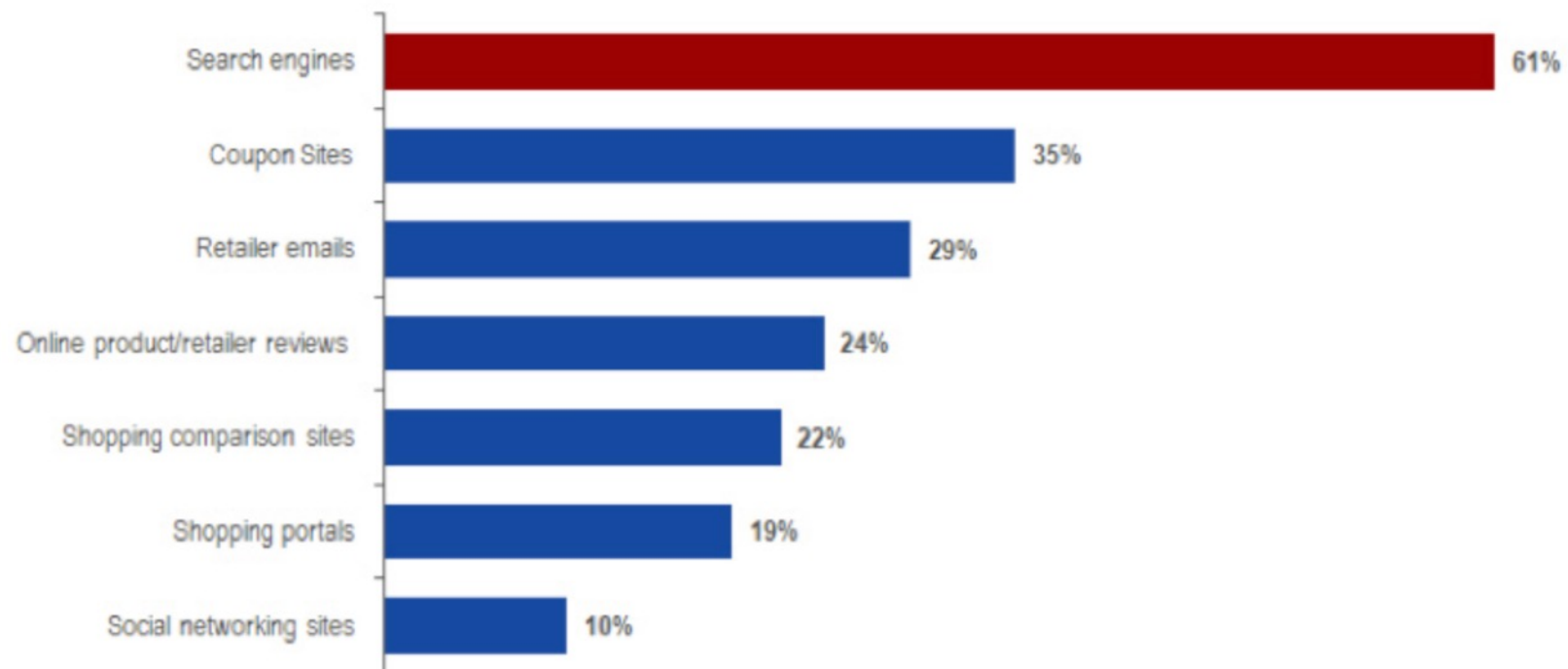
Data and information in perspective



Popularity of search engines



Popularity of search engines





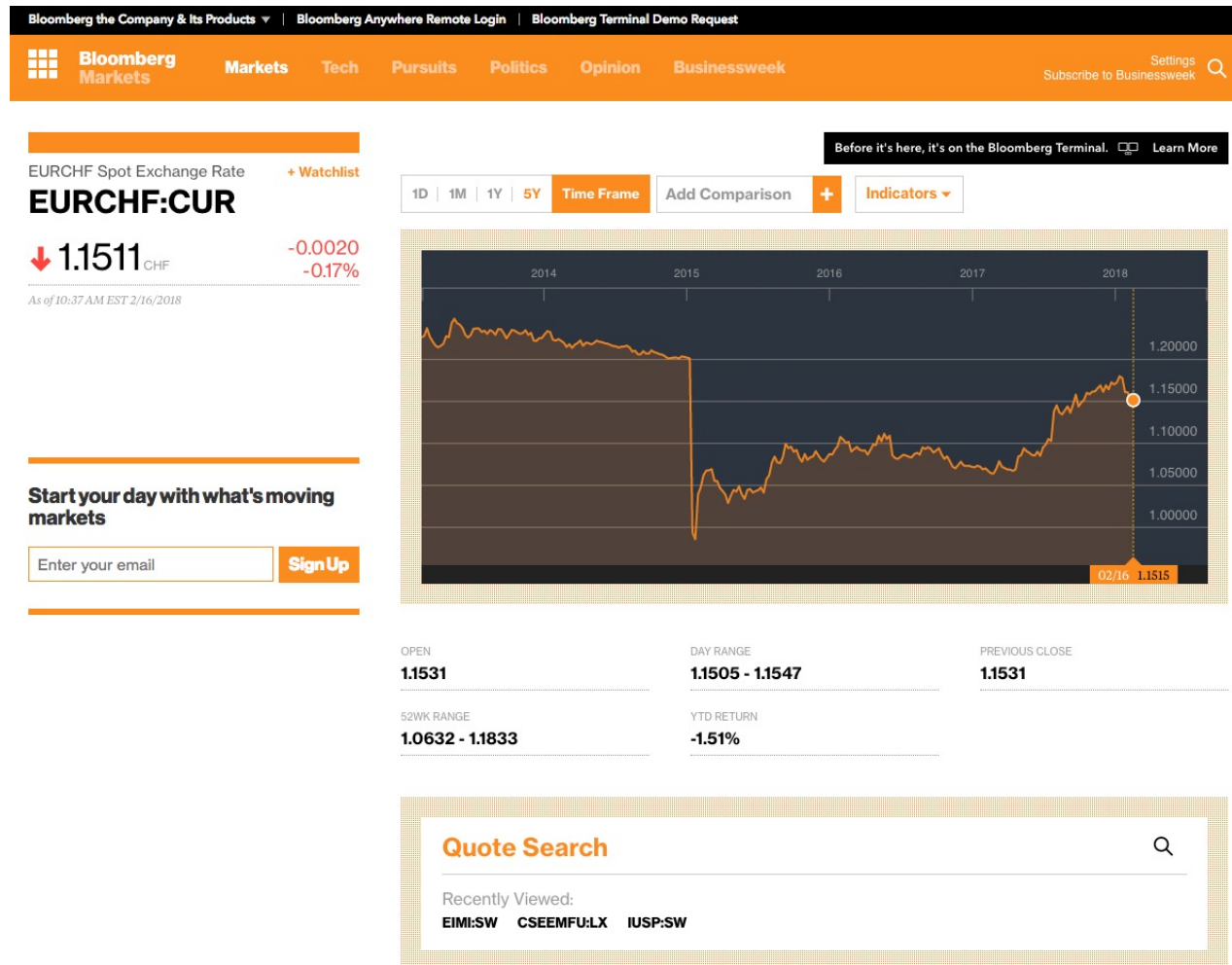
Volume

Repositories: Web



By Geni - Photo by user:geni, GFDL, <https://commons.wikimedia.org/w/index.php?curid=37841715>

Repositories: Bloomberg



Repositories: arXiv.org

 Cornell University
Library

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- High Energy Physics – Lattice ([hep-lat new](#), [recent](#), [find](#))
- High Energy Physics – Phenomenology ([hep-ph new](#), [recent](#), [find](#))
- High Energy Physics – Theory ([hep-th new](#), [recent](#), [find](#))
- Mathematical Physics ([math-ph new](#), [recent](#), [find](#))
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- Nuclear Experiment ([nucl-ex new](#), [recent](#), [find](#))
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Quantitative Finance

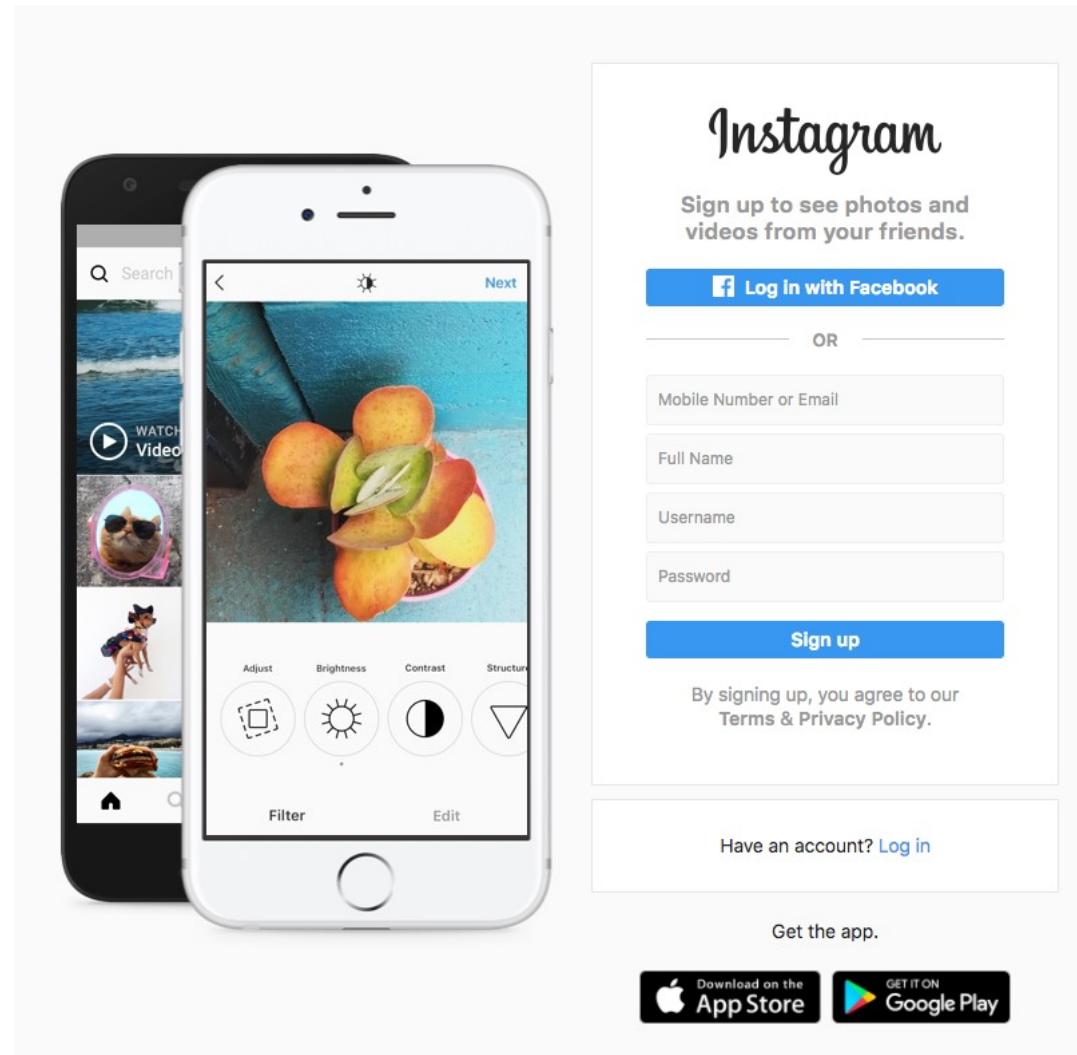
[Quantitative Finance](#)

Repositories: Amazon

The screenshot displays the Amazon.de homepage with the following elements:

- Header:** Amazon.de logo, search bar, and navigation links including "Halo! Lieferadresse wählen", "Alle Kategorien", "Mein Amazon.de", "Angebote", "Gutscheine", "Verkaufen", "Hilfe", "Amazon Prime", "Jetzt 30 Tage testen", "DE", "Hallo! Anmelden", "Mein Konto", "Testen Sie Prime", "Meine Listen", and "Einkaufswagen".
- Final Sale Banner:** A large banner for "FINAL SALE bis -70% Bekleidung, Schuhe, Taschen und mehr Fashion" featuring images of clothing, shoes, and bags.
- Amazon-Geräte:** A section titled "Amazon-Geräte" with a "Mehr" link, showing products like ECHO DOT, ECHO, ECHO SPOT, FIRE TV STICK, and FIRE 7.
- Spielzeug Bestseller:** A section titled "Spielzeug Bestseller" with a "Mehr" link, displaying popular toys such as EXIT, Uno, Gravitrax, and LEGO Duplo.
- Bücher Bestseller:** A section titled "Bücher Bestseller" with a "Mehr" link, showing best-selling books like "FEUER UND EIS" by Michael Wolff and "SCHLANK!" by Dr. med. Anne Fleck.
- Gift Cards:** A section titled "Amazon.de Geschenkgutscheine" with a "Was sind Cookies?" link, showing a gift card and a "Mit kostenloser Lieferung am nächsten Tag" offer.
- Final Fashion Sale:** A red banner at the bottom right for "Final Fashion Sale: bis -70% Rabatt" featuring a man and a woman in winter clothing.

Repositories: Instagram



Three paramount factors

Capacity

Throughput

Latency

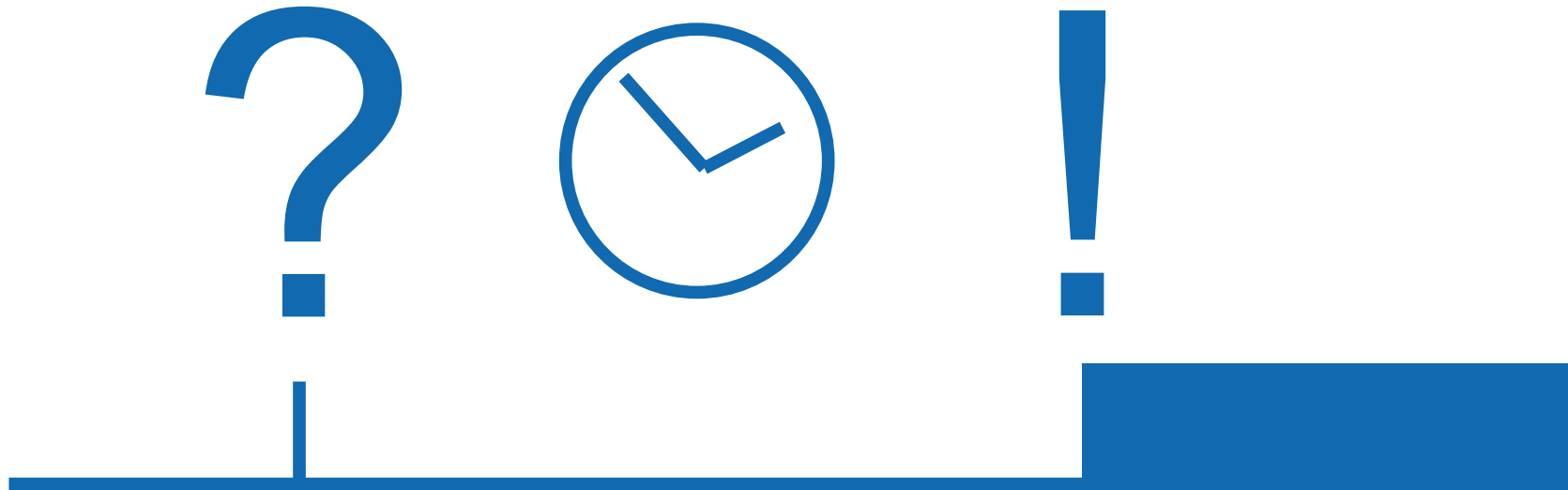
Capacity



Throughput



Latency



1956: IBM RAMAC 350



Capacity: 5 MB

Throughput: 12.5 kB/s

Latency: 600 ms

Dimensions:
1.7x1.5x0.7 (m)

1956: IBM RAMAC 350



Wikipedia
Deep silence

2022: Western Digital Ultrastar DC HC650



2022: Western Digital Ultrastar DC HC650



2022: Western Digital Ultrastar DC HC650



2022: Western Digital Ultrastar DC HC650



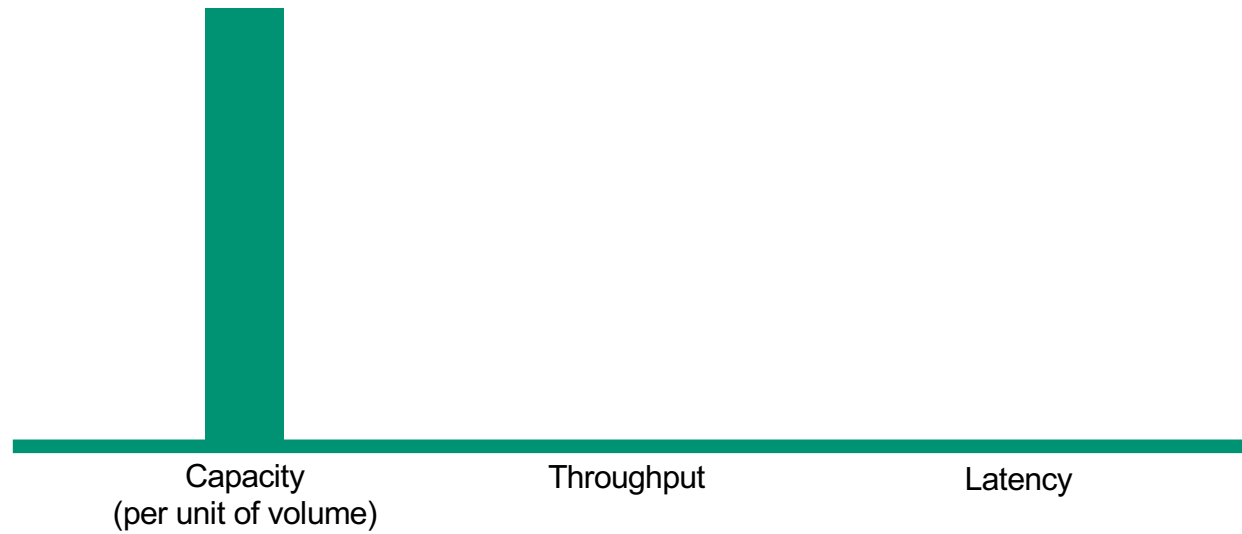
Capacity: 20 TB

Throughput: 250 MB/s

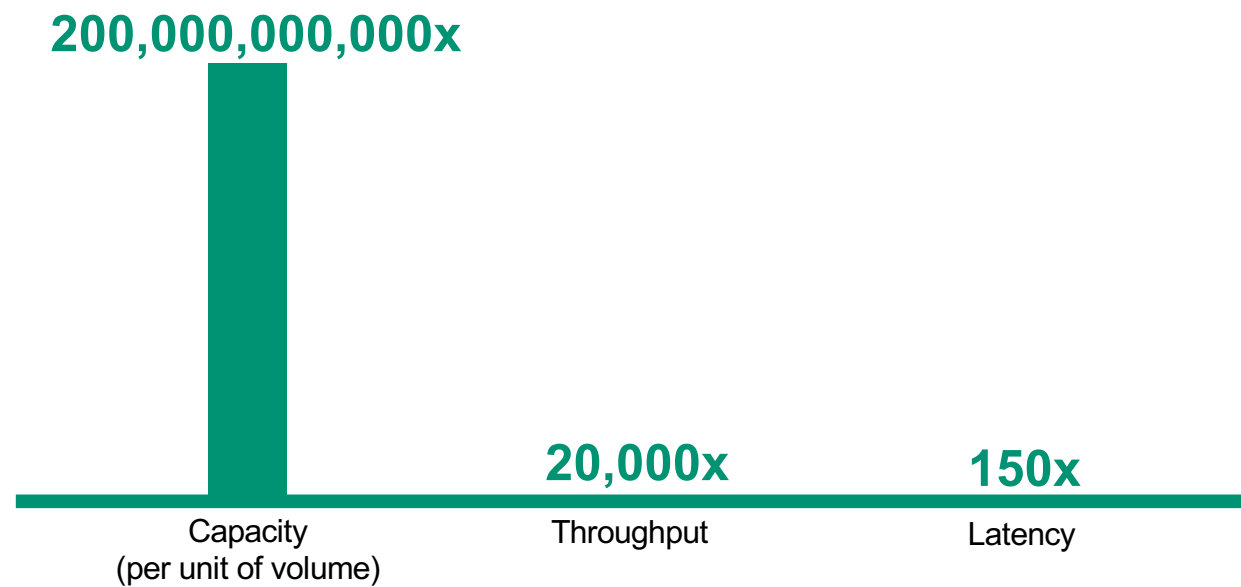
Latency: 4.16 ms

Dimensions:
101x147x26 (mm)

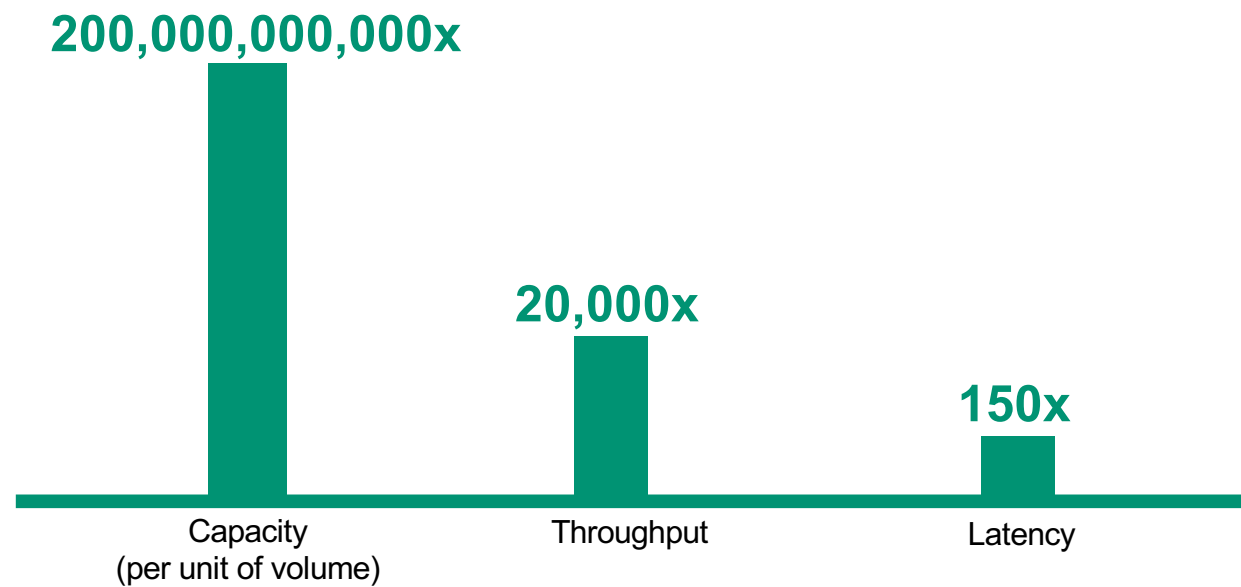
The progress made (1956-2022)



The progress made (1956-2022)



The progress made (1956-2022): Logarithmic

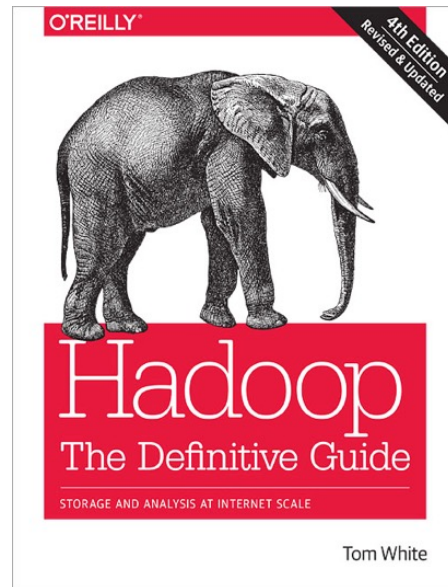


Growth

The difficulty seems to be, not so much that we publish unduly in view of the extent and variety of present day interests, but rather that **publication has been extended far beyond our present ability to make real use of the record**. The summation of human experience is being expanded at a prodigious rate, and the means we use for threading through the consequent maze to the momentarily important item is the same as was used in the days of square-rigged ships.

– Vannevar Bush, 1945.

Capacity: Example



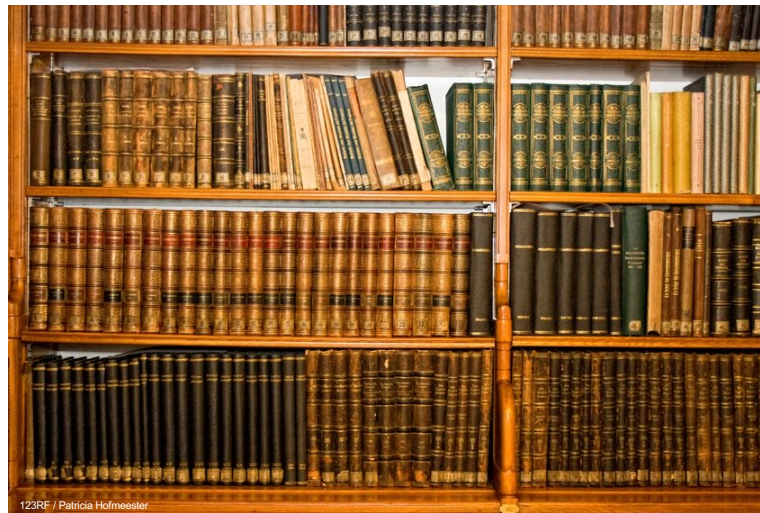
~ 600,000 words.

Throughput: Example



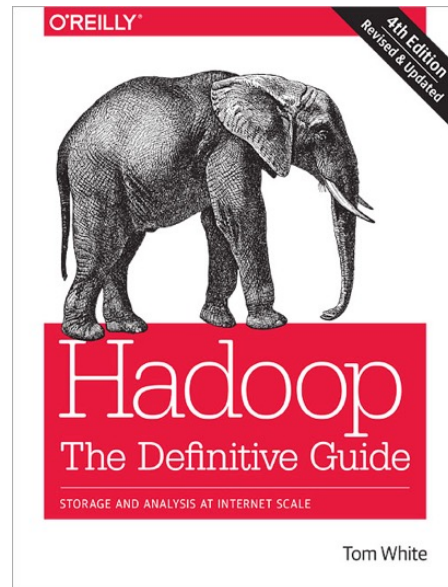
~1,000 word
per minute

Latency: Example



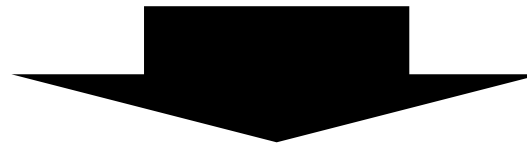
~ 1 minute to
stand up,
go to the shelf,
pick the book,
find the page.

2022 – Analogy with a book



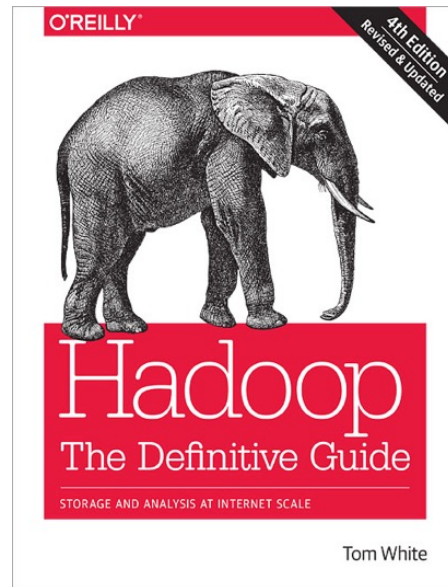
600,000 words

1,000 words per minute



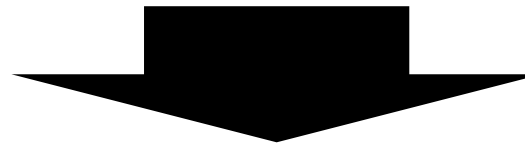
10 hours

2222 – Analogy with a book



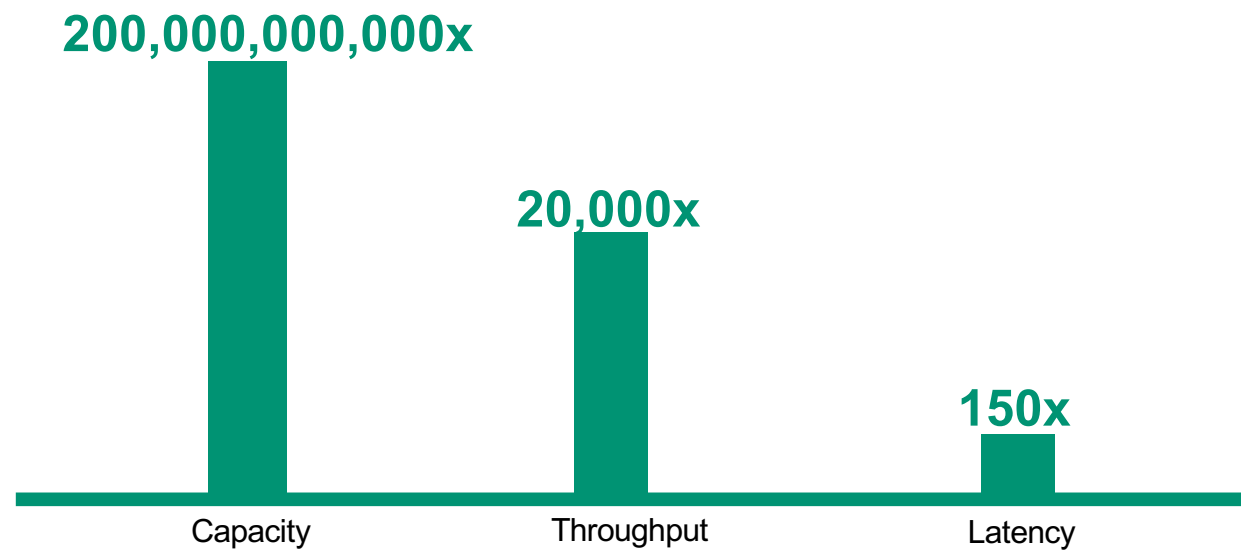
120,000,000,000,000,000 words

20,000,000 words per minute

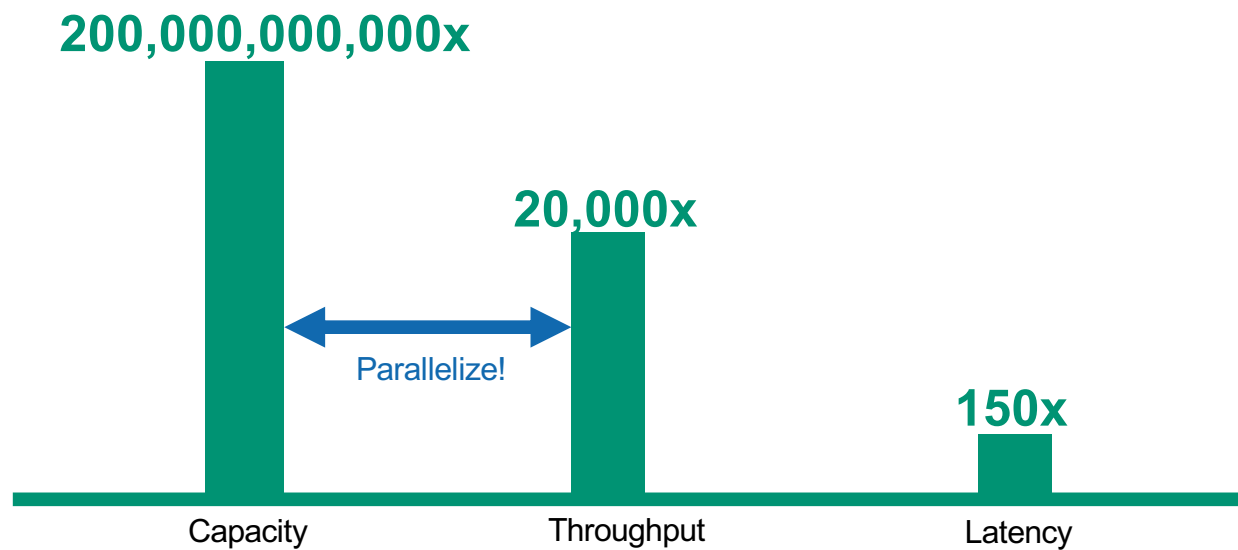


11,400 years

The progress made (1956-2022): Logarithmic



The progress made (1956-2022): Logarithmic



2222: 100,000,000 persons
could read it all in 10 hours.

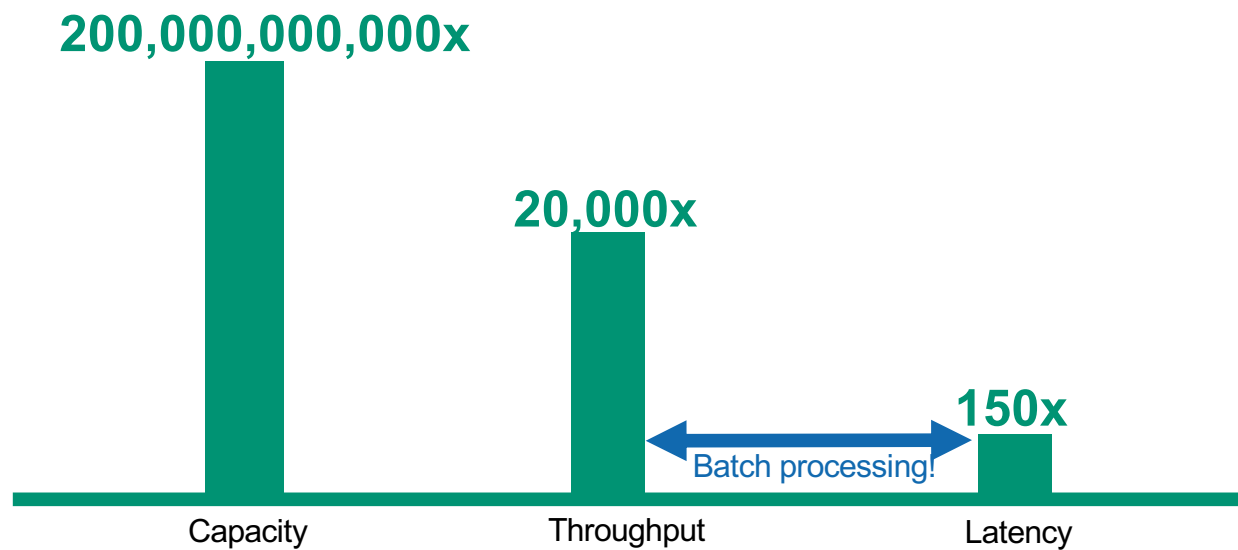


Data centers: clusters of machines (10,000s)



123RF /
blueringmedia

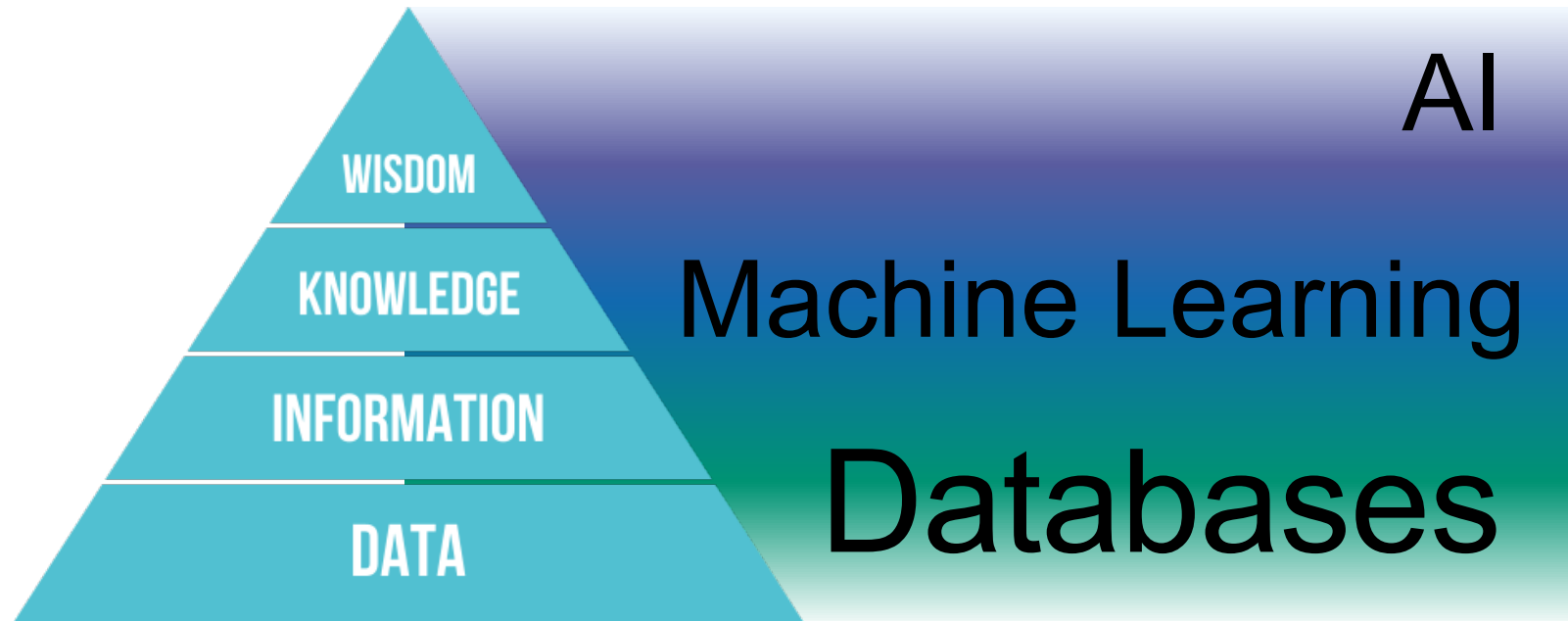
The progress made (1956-2022): Logarithmic



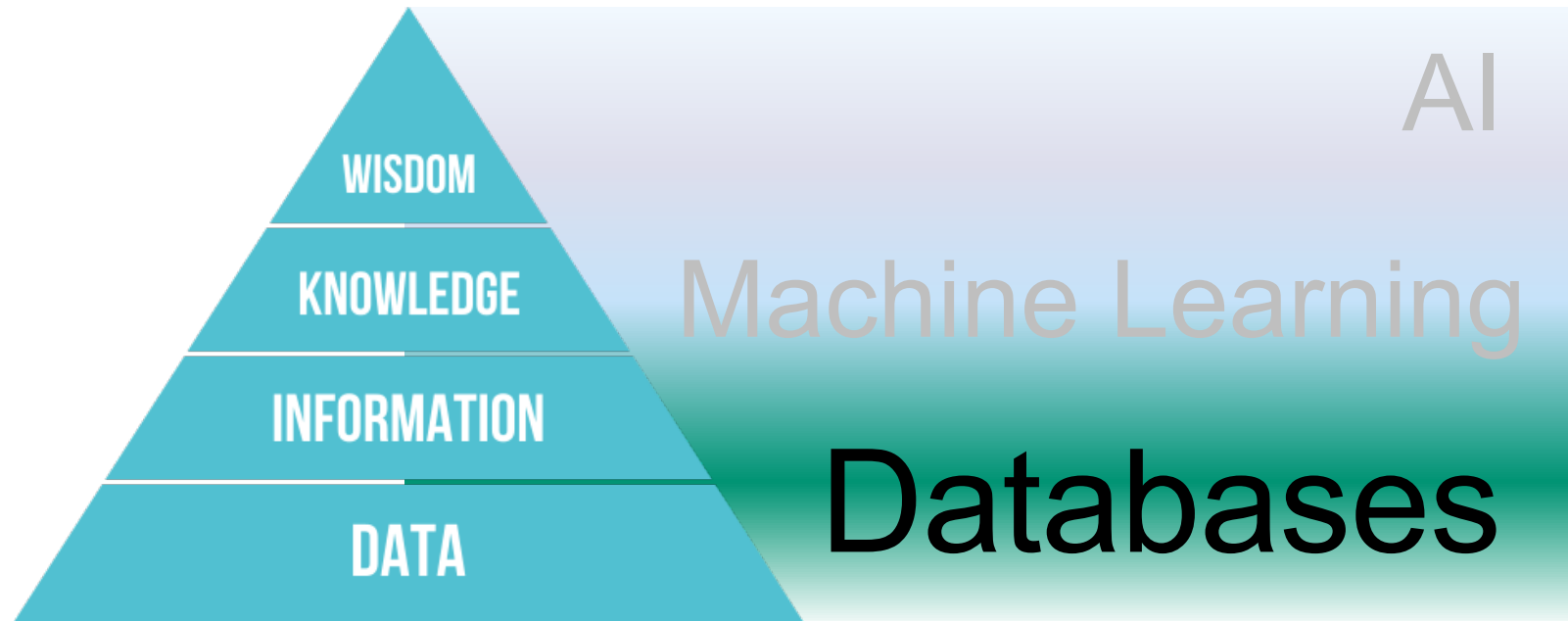
Lecture scope



Lecture scope



Lecture scope: no ML



Lecture Team



Ghislain Fourny



Andrea Lattuada
(TA)



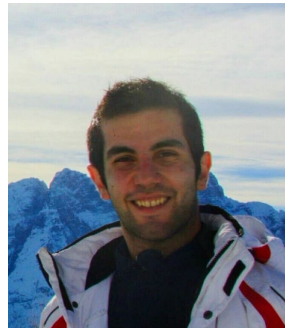
Abishek Ramdas
(TA)



Dan Graur
(TA)



**Christopher
Vogelsanger**
(TA)



Amir Joudaki
(TA)



Sonali Andani
(TA)



**Piriyakorn
Piriyatamwong**
(TA)

Lecture Overview

Introduction

Basics of Information Retrieval

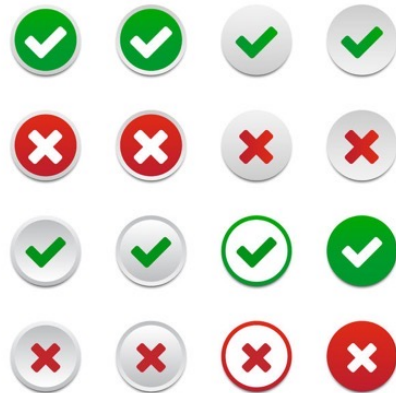


Lecture Overview

Introduction

Boolean queries

Basics of Information Retrieval



Lecture Overview

Introduction

Boolean queries

Term vocabulary and posting lists

Basics of Information Retrieval



Lecture Overview

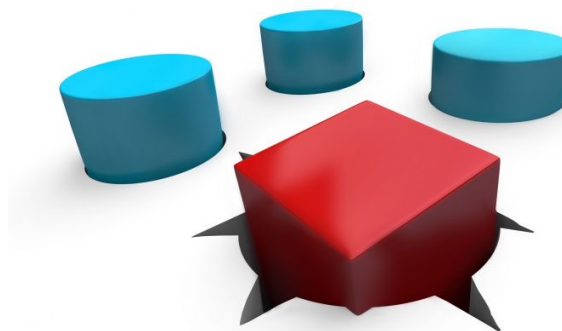
Introduction

Boolean queries

Term vocabulary and posting lists

Tolerant retrieval

Basics of Information Retrieval



Lecture Overview

Introduction

Boolean queries

Term vocabulary and posting lists

Tolerant retrieval

Scale up

Basics of Information Retrieval

Advanced topics



Lecture Overview

Introduction

Boolean queries

Term vocabulary and posting lists

Tolerant retrieval

Basics of Information Retrieval

Scale up

Index construction

Advanced topics



Lecture Overview

Introduction

Boolean queries

Term vocabulary and posting lists

Tolerant retrieval

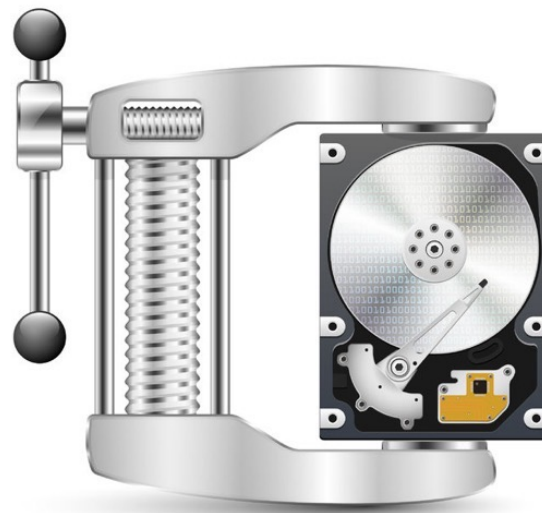
Basics of Information Retrieval

Scale up

Index construction

Index compression

Advanced topics



Lecture Overview

Introduction

Boolean queries

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Basics of Information Retrieval

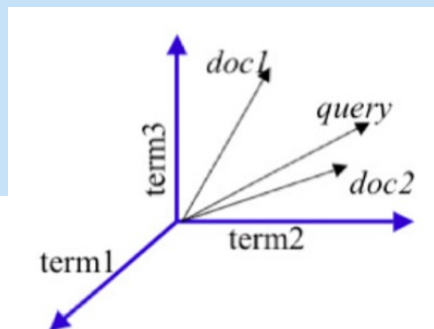
Scale up

Index construction

Index compression

Ranked retrieval

Advanced topics



Lecture Overview

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Term vocabulary and posting lists

Tolerant retrieval

Basics of Information Retrieval

Scale up

Index construction

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Ranked retrieval

Scoring optimization

Advanced topics



Lecture Overview

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Scale up

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Scoring optimization

Evaluation

Advanced topics



Lecture Overview

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Boolean queries

Term vocabulary and posting lists

Tolerant retrieval

Basics of Information Retrieval

Scale up

Index construction

Index compression

Ranked retrieval

Scoring optimization

Evaluation

Advanced topics

Probabilistic information retrieval

$$\text{sim}(d, q) = \sum_{w \in d} \left(\log \frac{P(w|R(q))}{1 - P(w|R(q))} - \log \frac{P(w|\bar{R}(q))}{1 - P(w|\bar{R}(q))} \right)$$

Alternate methodologies

Lecture Overview

Introduction

Boolean queries

Term vocabulary and posting lists

Tolerant retrieval

Basics of Information Retrieval

Scale up

Index construction

Index compression

Ranked retrieval

Scoring optimization

Evaluation

Advanced topics

Probabilistic information retrieval

Language models

Alternate methodologies



Lecture Overview

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Boolean queries

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Tolerant retrieval

Basics of Information Retrieval

Scale up

Index construction

Index compression

Ranked retrieval

Scoring optimization

Evaluation

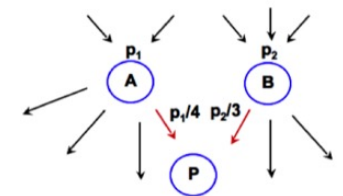
Advanced topics

Probabilistic information retrieval

Language models

Crawling the Web and link analysis

Alternate methodologies



What is expected

What is expected

Attendance of the **weekly lecture**
(2 hours/w Fridays 14-16)

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Attendance of the **exercise session**
(1 hour/w Fridays 16-17)

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Hands-on self-study, **read the books**,
play with technology

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Hands-on self-study, **read the books**,
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Passing the **written exam** in the
Summer session (180 minutes in August)

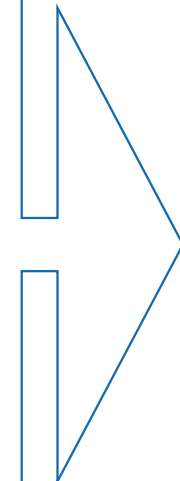
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Hands-on self-study, **read the books**,
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Passing the **written exam** in the
Summer session (180 minutes in August)



4
KP

Bonus points!

Bonus points!



0.25

Bonus points!

You SHOULD solve the weekly exercise sheets (theoretical, practical)

Bonus points!

You SHOULD solve the weekly exercise sheets (theoretical, practical)



We picked 3 weeks (on the webpage)

Bonus points!

You SHOULD solve the weekly exercise sheets (theoretical, practical)



We picked 3 weeks (on the webpage)



If you solved the exercises in 2 of these 3 weeks,

0.25 will be added to your exam grade

Bonus points!

You SHOULD solve the weekly exercise sheets (theoretical, practical)



We picked 3 weeks (on the webpage)

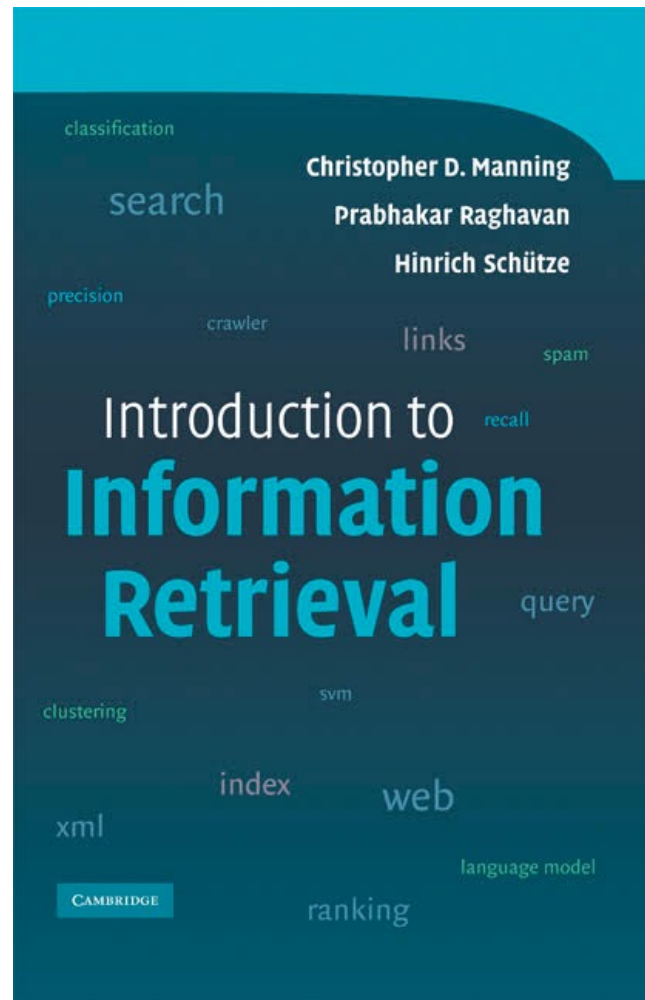


If you solved the exercises in 2 of these 3 weeks,

0.25 will be added to your exam grade

(Disclaimer: $6.00 + 0.25 = 6.00$)

Book



Programming assignments



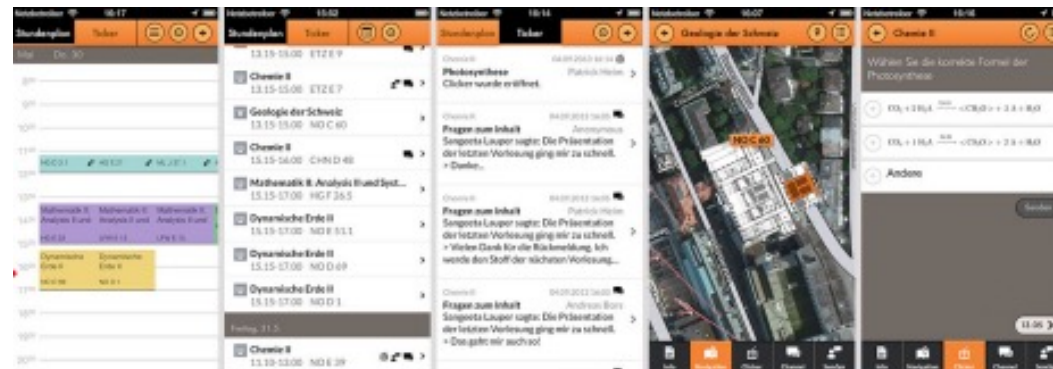
Poll

Go *now* to:

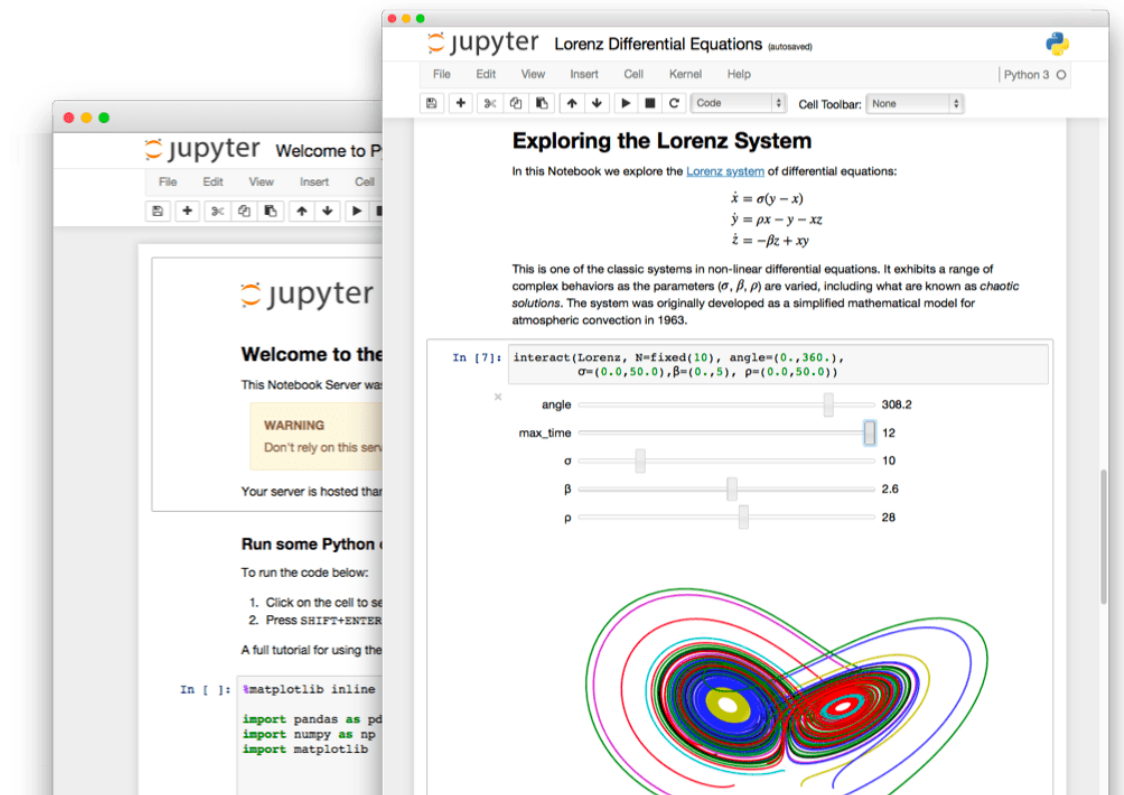
<https://eduapp-app1.ethz.ch/>



or install EduApp 3.x



Jupyter notebooks



Exercise session

Starts next week

4:15pm to 5:00pm

Before that, play with the



server

Register in mystudies now!

There are four in presence sessions, one of which allowing online participation.